Practice Plan, Aulas 1–2: Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 30 exercises of `Listas/pratica-aula-01-02.md`

October 2026

The 30 exercises are solved in the plan's order (foundations, exam level, advanced). Each is cited by book, number and page; the statement is in the book and is not reproduced. Books: Milgrom & Roberts (1992) (M&R), Cabral (2017), Varian (2014, with Calculus) and Mas-Colell, Whinston & Green (1995) (MWG). Page numbers are printed pages. Every computed number is checked by the Python scripts in this folder. Companion reading: the notes `Aula/notes-aulas-01-02/main.pdf`.

Contents

Tier 1 — Foundations **2**

1 M&R Food for Thought 1.1 (p. 18) **2**

2 M&R Food for Thought 1.2 (p. 18) **3**

3 Cabral 1.1 Competition and performance (p. 11) **5**

4 Cabral 3.9 Car parts (p. 62) **6**

5 M&R Food for Thought 4.1 (p. 119) **8**

6 Varian Ch. 32, Review Question 3 (p. 628) **9**

7 Varian Ch. 35, Review Question 1 (p. 687) **11**

8 Varian Ch. 35, Review Question 2 (p. 687) **13**

9 Cabral 5.1 Front yards (p. 117) **14**

10 Cabral 7.6 Hernan Cortez (p. 180) **16**

Tier 2 — Exam level **17**

11 M&R Food for Thought 1.3 (p. 18) **17**

12 Cabral 3.14 Intel (p. 63) **19**

13 Cabral 3.11 Smart car (p. 62) **20**

14 M&R Food for Thought 4.2 (p. 119) **22**

15 M&R Food for Thought 4.3 (p. 119) **24**

16 M&R Mathematical Exercise 2.1 (p. 53) **25**

17 M&R Mathematical Exercise 2.2 (p. 53) **27**

18 M&R Food for Thought 5.2 (p. 164) **28**

19 Cabral 7.14 Centipede (p. 182) **30**

20 M&R Food for Thought 5.1 (p. 163–164) **32**

21 M&R Analytical Problem 5.4 (p. 165) **34**

22 M&R Analytical Problem 5.5 (p. 165) **36**

23 M&R Food for Thought 2.5 (p. 52) **38**

Tier 3 — Advanced **40**

24 M&R Mathematical Exercise 4.2 (p. 120) **40**

25 Cabral 3.20 Productivity (applied) (p. 64) **42**

26 M&R Mathematical Exercise 2.3 (p. 53) **44**

27 M&R Mathematical Exercise 2.4 (p. 53) **46**

28 MWG Exercise 9.B.13 (p. 302–303) **48**

29 MWG Exercise 9.B.12 (p. 302–303) **50**

30 Cabral 13.10 Asset specificity and vertical integration (p. 346) **52**

Tier 1 — Foundations

1 M&R Food for Thought 1.1

Foundations · block A1

Source: M&R Food for Thought 1.1, page 18 (statement in the book; not reproduced here).

What is asked. Say which decisions in a fast-food chain are taken by headquarters and which by the outlet managers, explain why, and judge whether splitting the problems into coordination and motivation is a useful way to analyse the business.

Method. The organizing pair of the course: coordination (making plans fit, an information problem) versus motivation (getting people to carry them out, an interest problem) (M&R Ch. 1, p. 16; Ch. 2, "The Tasks of Coordination and Motivation", pp. 25–27). Authority goes where the information is, and incentives and delegated authority are complements (p. 17). Design variables are set centrally and details are left to local managers (Ch. 4, pp. 91, 114–115). Routines handle recurrent design problems (p. 92). Scale and scope economies (pp. 106–107).

Solution.

What the centre decides, and why. The menu and recipes, product specifications and portion sizes, food-safety and cleanliness standards, the operating manual, store layout and equipment, brand and national advertising, approved suppliers and central purchasing, and the launch of new products. These are *design variables* in the sense of p. 91. First, there is strong a priori information about the optimum: the chain's value is the promise that the product is the same in every outlet, so every outlet must match the standard. Second, a misfit is costly: one outlet that serves a different or unsafe product damages the brand everywhere. Many of these choices also carry scale economies (purchasing, advertising, product development; pp. 106–107) that no single outlet can capture. New products have *innovation attributes* (p. 92): testing a new item needs information and resources beyond any one outlet's experience.

What the outlet decides, and why. Crew hiring, training on the floor and supervision, shift schedules, the quantities ordered from approved suppliers, day-to-day maintenance, handling complaints and some local promotions. The information these need is local and changes fast: Friday's lunch rush, the weather, a local event, which worker is reliable. Sending it to headquarters would be slow and costly, and the information would be stale on arrival (Ch. 2, "The Need for Information", p. 26). Following p. 17, authority goes where the information is. The operating manual turns many recurring choices into *routines* (p. 92), so the manager solves them locally without asking the centre. In the language of p. 114, the central decisions set the parameters within which the local ones are made.

Coordination and motivation. The *coordination* problem is making thousands of kitchens produce an identical product at a predictable quality. It is solved mostly by central standards and routines. The *motivation* problem is that the manager's effort on cleanliness, speed and courtesy is hard to observe, and most of the gain from that effort goes to the brand, not to the outlet. An outlet serving travellers who rarely return bears little of the cost of shirking: the damage falls on all other outlets. The tools are an ownership stake in the outlet's profit (the franchise form makes the manager a residual claimant), plus inspections, standards backed by termination and required

input sources. Delegating local decisions pays only because the decision maker shares in the result (p. 17: incentives and delegation are complements).

Is the split useful? Yes, as a lens, but not as a way to sort decisions into two piles. Most decisions have both sides. Hygiene standards are centralized partly for coordination (uniformity) and partly for motivation: left alone, an outlet would under-invest in a brand it shares with others. Profit-based pay for a franchisee also creates a new motivation problem, since cutting costs below the standard raises its own profit at the brand's expense; the centre then monitors to protect the coordinated design. As M&R put it, motivation problems "can influence the effectiveness of coordination mechanisms" (p. 114). The good answer uses the split to ask two questions of each decision: who has the information, and whose interests does it touch?

Interpretation. This is the Aula 1 lesson on a small scale: the chain is a coherent system in which central standards, routines, local discretion and residual-claimant incentives support one another, the way GM's divisions, accounting and central office did (p. 17).

Verification. Limit cases. With no brand externality (independent restaurants), nearly everything would be decided locally and coordination through a central standard would lose its point. With perfectly observable effort, outlet managers could be salaried and closely directed, and ownership incentives would matter less. The observed split sits between the two, as the theory predicts.

Answer. The centre sets the design variables — menu, recipes, standards, layout, suppliers, branding, new products — because the optimum is known to require uniformity, misfits damage the whole brand, and they carry scale economies. Outlets decide staffing, scheduling, ordering and service, where information is local and fast-changing. The coordination/motivation split is a useful lens, but each decision has both aspects: standards are also a defence against free-riding on the brand, and outlet incentives must be paired with monitoring.

2 M&R Food for Thought 1.2

Foundations · block A1

Source: M&R Food for Thought 1.2, page 18 (statement in the book; not reproduced here).

What is asked. For an army in battle, identify which decisions are taken centrally and which are left to field commanders, state the principles behind the division, and judge whether the coordination/motivation split helps in analysing military organization.

Method. Coordination versus motivation (M&R Ch. 1, p. 16; Ch. 2, pp. 25–27). Authority goes to those with the information, and delegation is valuable only with aligned objectives (p. 17). Design attributes, synchronization and assignment problems (Ch. 4, pp. 91–92). Urgency plus central knowledge of the optimum favours central control (p. 92). Routines (p. 92). Central decisions set the parameters of local ones (pp. 114–116).

Solution.

What is decided centrally. The objective of the operation and the main line of effort; the *timing* of the attack; the boundaries between units; the commitment of reserves; the allocation of

artillery, air support and supplies among units; and the rules of engagement. Each is a design variable (p. 91). Timing is a *synchronization* problem: an assault on two axes, or infantry following a barrage, works only if everyone moves together. A unit that attacks early or late is destroyed piecemeal, which is the coxswain's problem of p. 91 with far higher stakes. Unit boundaries and fire support are *assignment* problems (pp. 91–92): each task needs exactly one unit, and the errors are a gap in the line (no one assigned) or fire on one's own troops (two assigned). As with the ambulance dispatcher, a central decision prevents duplication and delay, and a small error in the overall plan costs far less than a failure of fit. The centre also collects intelligence from many sources (reconnaissance, other sectors), which no field commander has. This is an innovation attribute (p. 92): the needed information is not available to any one unit.

What is left to field commanders. How to take the objective given the terrain; fire and movement; when to exploit an opening or fall back; how to react when the enemy does something unexpected. The information is local, perishable and observed only on the spot. Communication is slow and is destroyed by the battle itself, so routing it upward would make the decision both late and uninformed — the HBC's failure of over-centralization (p. 26) in its starkest form. Hence a familiar military principle: the centre issues the *intent* (what and why, by when) and leaves the *method* (how) to the subordinate. This is David Packard's rule quoted on p. 116: agree on the objective, then "turn everybody loose". Drills and standard procedures are routines (p. 92) that let units act coherently even when contact with the centre is lost.

Principles. (1) Centralize what has design attributes: timing, boundaries, assignments, the overall plan. (2) Decentralize what depends on local, fast-changing information. (3) The more urgent and disrupted the communication, the more must be decided below, within parameters set above (p. 114).

Coordination and motivation. The split is useful, and motivation in an army is extreme: the organization asks people to risk their lives, so individual interest (survival) and the organization's goal (winning) diverge sharply. Output is produced by teams and is hard to measure, so pay for performance is nearly useless. Armies rely instead on selection and training, discipline and punishment, unit cohesion and loyalty, honours, and promotion based on careers. But the two problems interact. Delegation to field commanders pays only if they share the objective (p. 17): an army that cannot trust its officers' competence or loyalty must centralize more and accept slower, less informed decisions. Motivation also shapes coordination through reporting. Commanders who are judged on success have reasons to overstate progress, which corrupts the information the centre plans with.

Interpretation. The army shows the course's two principles at their limit. Central control of design variables buys fit, local control buys speed and information, and the boundary between them is set by the cost of misfit, the speed of change and the trust that incentives and selection create.

Verification. Cross-check against the opposite extreme. A fully centralized army, with every tactical move approved above, fails on information and speed. A fully decentralized one, with each unit choosing its own objective and timing, fails on synchronization and assignment. Both fail in the predictable ways of p. 93 (central commands: quick but "insensitive to local knowledge"; decentralized: duplication and delay).

Answer. The centre sets objectives, timing, unit boundaries, reserves, fire support and supply allocation — design variables with synchronization and assignment attributes, where misfits are catastrophic. Field commanders choose the methods, using local, perishable information within the centre's intent. The coordination/motivation split is useful, but the two interact: delegation is viable only with trained, loyal and motivated commanders, and incentive problems distort the information that coordination needs.

3 Cabral 1.1 Competition and performance

Foundations · block A1

Source: Cabral 1.1 Competition and performance, page 11 (statement in the book; not reproduced here).

What is asked. Cabral reports a UK firm-level finding: holding inputs fixed, output rises with the number of competitors and falls with market share and concentration. Relate it to the ideas of Chapter 1.

Method. Chapter 1 lists three costs of market power: allocative inefficiency, productive inefficiency ("the best of all monopoly profits is the quiet life") and rent seeking (Cabral §1.2, pp. 6–7), against the Schumpeterian defence of market power (p. 7). The channels come from later chapters: product-market discipline of managers (§3.3, p. 53), X-inefficiency (§5.3, p. 103) and competitive selection (§4.2, p. 75). Course link: M&R's "incentives matter" pattern (Ch. 1, p. 16).

Solution.

(1) What the finding measures. "Output for given inputs" is productivity: in the language of §3.1 (pp. 39–40) the result says that total factor productivity is higher where competition is stronger. That is a statement about *productive* efficiency, the second of Cabral's three inefficiencies, not about prices. A monopolist that produces at minimum cost still causes allocative inefficiency (a price above marginal cost); the UK evidence says that, on top of this, firms with market power do not produce at minimum cost.

(2) The "quiet life" reading. Chapter 1 argues that a firm protected from competition "does not need to be bothered" and has weaker incentives to be cost efficient (p. 6), with European versus North-American airlines as the example. The UK result is the cross-section version of the same claim: the more competitors (and the smaller the firm's share and the industry's concentration), the closer the firm is to the efficient frontier. Section 5.3 states the same evidence explicitly: firms in more competitive markets are more productive, and most of the gap is X-inefficiency (p. 103).

(3) Why competition raises productivity: three channels.

(a) *Discipline through survival.* Under intense competition a manager who does not push for efficiency raises the chance that the firm fails and he loses his job (§3.3, p. 53; formalized in Exercise 3.19).

(b) *Yardstick information.* Rivals' results reveal what is achievable. In a monopoly the manager can blame poor results on industry conditions; with competitors that excuse fails (p. 53). Competition lowers the owners' information disadvantage, which is the agency problem of §3.3.

(c) *Selection.* Competitive markets push out low-productivity firms (§4.2, p. 75), so the surviving average is higher where competition is stronger.

Channels (a) and (b) operate *inside* the firm: the market around the firm changes the incentives of the people in it. That is the course's point: an organization's incentive system and its competitive environment are not separable. M&R's Hudson's Bay Company case is the historical version: a firm with a royal monopoly charter and weak internal incentives was beaten by the North West Company, and competed effectively only after it copied the rival's strategy and structure (Ch. 1, pp. 7, 16).

(4) The counter-argument. The Schumpeterian view (p. 7; §15.1) holds that market power funds innovation, so a static productivity gap could coexist with faster dynamic progress in concentrated industries. The UK evidence is about productivity levels; it does not settle the dynamic question.

Interpretation. Topic 1 asks whether organization matters. This exercise shows one reason it varies across firms: the same internal structure performs differently depending on how much pressure the market applies. Competition is a substitute for explicit incentive contracts (it supplies both the stick and the yardstick), so a firm with market power must provide by design the discipline that a competitive firm gets from its rivals.

Verification. Two checks on the reading. *Reverse causality:* if efficient firms simply grew large (an efficiency explanation of concentration), the cross-section would show a *positive* link between market share and productivity. The UK data show a negative one, so the pure efficiency story cannot explain them. *Omitted variables:* concentration is endogenous (industries differ in scale economies and demand), so the correlation is evidence consistent with channels (a)–(c), not proof of causality, the same caveat Cabral attaches to the management-quality regressions (§3.5, p. 58).

Answer. The finding is evidence that market power causes *productive* inefficiency (the "quiet life", p. 6; X-inefficiency, p. 103), on top of allocative inefficiency. Competition disciplines managers (survival threat), supplies yardsticks (less asymmetric information) and selects efficient firms (§3.3, §4.2). Hence incentives inside a firm depend on the market around it. The correlation is suggestive, not causal, and says nothing against the Schumpeterian dynamic argument.

4 Cabral 3.9 Car parts

Foundations · block A2

Source: Cabral 3.9 Car parts, page 62 (statement in the book; not reproduced here).

What is asked. Of two parts of a car's taillight, the plastic exterior cover and the light bulb, say which a carmaker is more likely to make in-house, and why.

Method. The make-or-buy trade-off of §3.4 (pp. 54–56): vertical boundaries balance *investment incentives* (specific assets and the hold-up problem) against *performance incentives* (an outside supplier who can be replaced works harder), with intermediate forms such as long-term supplier relations (p. 55). The mould for a car part is Cabral's own example of a specific asset (§13.1, p. 338). Course cases: GM's integration and Toyota's outside suppliers (M&R Ch. 1, pp. 4–6).

Solution.

Compare the two parts on the dimensions that decide make or buy.

	Exterior cover	Light bulb
Asset specificity	high: shape, colour and fit are unique to one model; the mould is worthless for anything else	low: a standard part sold to many carmakers and in the aftermarket
Number of suppliers ex post	one (whoever holds the mould)	many
Scale economies	limited to one model's volume, which the carmaker can reach itself	large: a specialist pools demand from the whole industry
Design coordination	tight: must match the body, the lamp housing and the styling	minimal: fits a standard socket
Supplier know-how	moulding plastics, close to what a carmaker already does	specialized lighting technology

The bulb is bought. It is a standard part: no investment is specific to one buyer, so no hold-up arises; competing suppliers keep prices near cost and the threat of switching keeps performance high (p. 55); and a specialist that serves the whole industry reaches scale economies that no single carmaker could. Making it in-house would give up the market's performance incentive and its scale for no gain.

The cover is the candidate for making in-house. Producing it requires a model-specific mould. Once an outside supplier has sunk that investment, the carmaker can squeeze the price (the supplier has no other customer for the mould); once the carmaker has committed to the supplier's mould in the middle of a model run, the supplier can raise the price (no one else can supply the part at short notice). Either side can hold up the other, which is Cabral's coal-mine and power-plant logic (p. 55) and the mould example of §13.1 (p. 338). Ownership removes the bargaining between the two sides. Close design coordination with the body and the small extra know-how needed point in the same direction.

Qualification: intermediate forms. "More likely in-house" is a comparative statement. The hold-up problem can also be handled without full ownership: a long-term relation with a few trusted suppliers who are willing to invest in buyer-specific assets (the Japanese supplier system, p. 55), or an arrangement where the carmaker owns the specific mould while the supplier runs it, which takes the specific asset off the supplier's books and lets the carmaker move the mould to another supplier if the price is raised. This is the GM-versus-Toyota contrast of Aula 1: GM's volumes let it run efficient parts plants and it integrated (M&R p. 5), while Toyota, too small to reach efficient scale in parts, used few outside suppliers in long-term relations for complex components (pp. 4–6).

Interpretation. The make-or-buy answer depends on the transaction, not on the industry: the same carmaker should buy the commodity and make, or govern closely, the specific part. Specificity creates hold-up, which favours integration; standardization makes market discipline available, which favours buying. Item 30 (Cabral 13.10) puts numbers on the hold-up side.

Verification. Limit cases. If the cover were a standard shape used across many models and makers, its mould would stop being specific and the cover would move to the "buy" column. If a

bulb were a custom design for one model (a model-specific LED unit, say), it would move towards the "make or govern closely" column. The prediction follows the specificity of the asset, not the physical part.

Answer. The plastic exterior cover is more likely made in-house: its model-specific mould is a specific asset, so outsourcing exposes both sides to hold-up, and design coordination with the body is tight. The bulb is a standard part with many suppliers and large scale economies, so it is bought and market competition disciplines the supplier (§3.4, p. 55). Long-term supplier relations (Toyota) are the intermediate solution.

5 M&R Food for Thought 4.1

Foundations · block A1

Source: M&R Food for Thought 4.1, page 119 (statement in the book; not reproduced here).

What is asked. Explain whether efficiency can justify the switch of market economies to central planning of scarce resources in the two world wars, and what other (non-efficiency) reasons support it.

Method. The three criteria for comparing coordination systems: efficiency with perfect information, communication, brittleness (M&R Ch. 4, pp. 93–94). Prices versus quantities and Equation 4.1 (pp. 94–98). Design and innovation attributes (pp. 91–93) and planning with design attributes (pp. 103–106). The failure of prices under scale economies (summary, p. 117). M&R note that planners in the most market-oriented economies directed steel, rubber, sugar and meat in the Second World War (p. 89).

Solution.

Efficiency reasons.

(i) *A sudden, huge change with innovation attributes.* War replaces the peacetime pattern of output with a new one that nobody in the market has experience of: ships, aircraft, munitions in enormous quantities. The information on what is needed sits with the military and the government, not with dispersed buyers and sellers. A price system would have to grope its way to an entirely new equilibrium (p. 103), and this takes time a war does not allow. Problems whose optimum depends on information that operating units lack are badly solved by decentralized prices (p. 92).

(ii) *Design attributes on a national scale.* War production is a giant synchronization and fit problem. Aircraft need engines, aluminium, fuel, trained crews and shipping at the same time and place. An operation needs everything in the same week. A priori the pattern of the solution is known (the parts must match), and a misfit is far more costly than a slightly wrong overall scale (p. 91, p. 105). Announcing quantities and dates is then both cheaper in communication and less brittle than prices (pp. 104–106).

(iii) *Brittleness.* By Equation 4.1 (p. 97), the loss under a price control relative to a quantity control equals the slope of marginal benefit over the slope of marginal cost. For critical war goods the marginal benefit is very steep: falling short of the required quantity of ammunition or fuel is disastrous, and a surplus is worth little. Marginal cost, by contrast, is fairly flat once mass production is set up. A steep marginal benefit relative to marginal cost is exactly

the case in which a quantity directive loses less than a price when the planner's estimates are wrong (pp. 96–98).

(iv) *No market to price against.* The government is the single buyer of munitions, and the new mass-production lines have large scale economies, so market-clearing prices may not exist or may not guide output efficiently (p. 117).

Other reasons.

- *Distribution and morale.* With civilian goods cut back, market prices would ration sugar or meat by ability to pay. Coupon rationing spreads the sacrifice evenly, which matters for consent to the war. This is a distributional argument, not an efficiency one.
- *Inflation and windfall gains.* A sharp rise in prices transfers large rents to the owners of scarce inputs and raises the government's own procurement bill. Price controls with rationing suppress both, though at the cost of shortages and black markets.
- *Urgency and authority.* War concentrates decision power and gives the government a single, widely shared objective, which reduces the political cost of command.

The other side. Central planning keeps its usual costs: it ignores local information about costs, it invites the motivation problems of planned economies (the ratchet effect, p. 14), and it breeds black markets. That is why even wartime economies planned only the critical sectors, left many decisions to firms and consumers, and dismantled controls after the war. The efficiency case is strongest for the military-industrial core (design and innovation attributes, steep marginal benefit) and weakest for consumer goods, where the case for rationing is mainly distributional.

Interpretation. Which coordination system is best depends on the problem, not on ideology. The same criteria that favour prices for routine resource allocation (Hurwicz, pp. 100–103) favour quantities and central direction when a problem has design and innovation attributes, urgency and a steep cost of shortfall.

Verification. Consistency check with Table 4.1 (p. 95). There, with the planner wrong about costs, a quantity directive loses 1 unit of net benefit and a price directive loses 15. The table is the case of a marginal benefit that is steep relative to marginal cost — the wartime case. In peacetime, with flat marginal benefit (many substitutes) and steep marginal cost, Eq. 4.1 reverses the ranking, consistent with returning to prices after the war.

Answer. Yes. War creates an abrupt change with innovation attributes, a national problem with design attributes (synchronization and fit of production), a steep marginal benefit of critical goods that makes quantity directives less brittle than prices (Eq. 4.1), and monopsony and scale economies that undermine price guidance. Non-efficiency reasons — equitable rationing for morale, curbing inflation and windfall profits, urgency — reinforce the case, which is strongest for military production and mainly distributional for consumer goods.

6 Varian Ch. 32, Review Question 3

Foundations · block B1

Source: Varian Ch. 32, Review Question 3, page 628 (statement in the book; not reproduced here).

What is asked. Decide whether knowing the contract curve of an exchange economy is enough to know where trade between the two consumers will end up, and justify the verdict.

Method. Pareto efficiency in the Edgeworth box and the contract curve (Varian §32.3, pp. 605–607); the role of the endowment and of the trading mechanism (competitive equilibrium, take-it-or-leave-it offers, Nash bargaining); the course notes, Part II, sections "The question: is bargaining indeterminate?" and "Dividing the gains: the Nash bargaining solution".

Solution. Verdict: false. The contract curve tells us *which kind* of point trade ends at, not *which* point.

(a) What the contract curve is. In a two-person, two-good exchange economy with total endowment $(\bar{\omega}_1, \bar{\omega}_2)$, an allocation is Pareto efficient when no reallocation makes one consumer better off without making the other worse off. At an interior efficient allocation the indifference curves are tangent,

$$MRS_{12}^{A}(x^A) \;=\; MRS_{12}^{B}(\bar{\omega} - x^A),$$

because crossing curves leave a lens of mutually preferred allocations (Varian p. 606). The contract curve (Pareto set) is the locus of all such points; it typically runs from A's origin to B's origin, and it is defined by preferences and totals alone.

(b) What it does tell us. If trade is voluntary and the parties keep trading until no mutually advantageous trade remains, the final allocation is on the contract curve: a point off the curve would not be a "final contract" (Varian p. 606). Voluntary trade also requires that each consumer end up at least as well off as at the endowment ω. So trade ends in the segment of the contract curve inside the lens through ω (Varian p. 607; this segment is the core of the two-person economy). This is exactly Varian's answer (p. A27): trade ends "somewhere on the curve; however, we don't know where." We agree, and add what decides the point.

(c) What it does not tell us. Two things, neither of which is in the contract curve:

(i) *The endowment.* A different ω selects a different segment.

(ii) *The mechanism.* Within the segment, the point is set by how trade happens: a competitive (Walrasian) equilibrium, a take-it-or-leave-it offer by one side, or a bargaining rule that splits the gains according to bargaining power and patience.

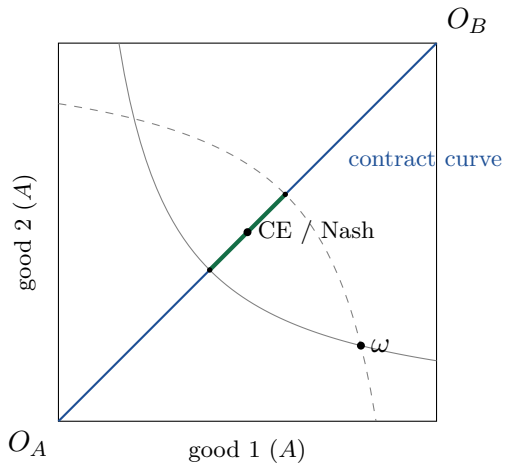
Example. Let $u_A = \ln x_1^A + \ln x_2^A$, $u_B = \ln x_1^B + \ln x_2^B$, totals $(10, 10)$ and endowment $\omega^A = (8, 2)$, $\omega^B = (2, 8)$. Tangency gives $x_2^A / x_1^A = (10 - x_2^A)/(10 - x_1^A)$, i.e. the contract curve is the diagonal $x_1^A = x_2^A = t$. Individual rationality on the diagonal:

$$u_A:\; t^2 \geq 8 \cdot 2 = 16 \;\Rightarrow\; t \geq 4, \qquad u_B:\; (10-t)^2 \geq 16 \;\Rightarrow\; t \leq 6.$$

So trade ends somewhere on the segment $t \in [4, 6]$, and the mechanism picks the point:

Mechanism	Final allocation of A
A makes a take-it-or-leave-it offer	$t = 6$ (B indifferent to ω)
B makes a take-it-or-leave-it offer	$t = 4$ (A indifferent to ω)
Competitive equilibrium ($p_1/p_2 = 1$, A's income 10)	$t = 5$
Symmetric Nash bargaining	$t = 5$

All four outcomes are on the same contract curve; they differ in who gets the gains from trade.



Solid grey: A's indifference curve through ω; dashed: B's. Green: the part of the contract curve both prefer to ω, where voluntary efficient trade ends.

(d) A caveat in the other direction. Even "somewhere on the curve" presumes that trade exhausts the gains. With private information about valuations, no bargaining procedure may reach the contract curve (M&R Ch. 5, pp. 141–143; notes, Part II, "Private information"). The contract curve is then the set of *possible* efficient outcomes, not a prediction.

Interpretation. This is the exchange-economy version of M&R's separation (pp. 36–38): efficiency restricts *what* is done, while bargaining power, patience and the initial position decide *who gets what*. Efficiency is a property of the outcome; the division needs a theory of the bargaining process (Nash, Rubinstein), which is why the course moves on to bargaining after the value maximization principle.

Verification. At $t = 4$, $u_A = 2\ln 4 = \ln 16 = u_A(\omega)$; at $t = 6$, $u_B = 2\ln 4 = u_B(\omega)$. With $p_1 = p_2 = 1$, A's income is $8 + 2 = 10$ and Cobb–Douglas demand spends half on each good: $x^A = (5, 5)$, $x^B = (5, 5)$, markets clear. Symmetry of the Nash product $(2\ln t - \ln 16)(2\ln(10 - t) - \ln 16)$ around $t = 5$ gives $t = 5$ (`check_C.py`).

Answer. False. The contract curve is the set of efficient allocations; voluntary efficient trade ends on its segment inside the lens through the endowment, but the point within that segment depends on the endowment and on the trading mechanism (in the example, anywhere in $t \in [4, 6]$: 4 or 6 under take-it-or-leave-it offers, 5 under competition or symmetric Nash).

7 Varian Ch. 35, Review Question 1

Foundations · block B1

Source: Varian Ch. 35, Review Question 1, page 687 (statement in the book; not reproduced here).

What is asked. Judge the claim that clearly defining property rights usually removes the problem of externalities, saying what such a delineation achieves and what it leaves open.

Method. Externalities as missing or ill-defined property rights (Varian §35.1, pp. 667–670); quasilinear preferences and the Coase theorem (Varian §35.2, pp. 670–672); M&R's value maxi-

mization principle and Coase theorem with their hypotheses (pp. 35–39); bargaining costs, private information and many parties (M&R Ch. 5, pp. 140–149).

Solution. Verdict: true as a statement about efficiency in small-numbers, low-cost bargaining; false as a general policy claim. Varian's own answer (p. A29) is "True" with a distributional caveat. We agree with it and make the qualifications explicit.

(a) What delineation fixes. An externality is an effect of one agent's action on another's utility or production that is not priced. Varian's diagnosis is that "the practical problems with externalities generally arise because of poorly defined property rights" (p. 670): if the smoker believes he may smoke and the roommate believes she is entitled to clean air, nobody can buy or sell the right, and the level of smoke is set by whoever acts. Once the right is assigned and enforceable, it becomes an endowment that can be traded like money or bread. Starting from that endowment, voluntary trade moves the parties to the contract curve, where

$$MRS^{A}_{\text{smoke, money}} \;=\; MRS^{B}_{\text{smoke, money}},$$

so the level of the externality is Pareto efficient *whichever* party holds the right (Varian p. 669, points X and X' of Figure 35.1).

(b) What delineation does not fix.

(i) *Distribution.* Assigning the right is assigning an endowment. The holder is paid to give it up rather than paying to obtain it, so the final money holdings differ across assignments (Varian p. A29; item 8 computes the difference).

(ii) *The efficient level itself, when there are wealth effects.* Without quasilinearity the efficient amount of the externality depends on who holds the right (Varian p. 670). Example: a plant gains $G = 40$ from dumping into a river; a resident with wealth $w = 100$ has $u = \ln x + a\,y$, $y = 1$ if the river is clean, with $a = \ln 1.5$. Then

$$\text{WTP} = w(1 - e^{-a}) = 100\big(1 - \tfrac{2}{3}\big) = 33.33, \qquad \text{WTA} = w(e^{a} - 1) = 100(1.5 - 1) = 50.$$

If the plant holds the right, the resident can offer at most $33.33 < 40$: the river is polluted. If the resident holds it, the plant would have to pay at least $50 > 40$: the river stays clean. Both outcomes are efficient given their endowment, yet the *real* outcome differs. Delineation still delivers efficiency, but "the" efficient level no longer exists independently of the assignment (this is the gap between value maximization and efficiency under wealth effects, M&R n. 10, p. 37).

(iii) *Transaction costs.* The Coase theorem assumes that the parties "bargain to an efficient agreement" (M&R p. 38). That fails when

- there are many parties (air pollution over a city): each victim free rides on the others' payments, and the chance that anyone volunteers stays bounded away from one even when the project is surely value maximizing (M&R pp. 145–147);
- valuations are private: no bargaining procedure may reach the efficient trade (M&R pp. 141–143);
- measuring and enforcing the right is costly (diffuse emissions, noise at 3 a.m.), or bargaining itself absorbs resources (M&R pp. 147–149);
- parties are wealth constrained and cannot pay the compensation that efficiency requires (M&R p. 39).

In these cases, rights alone leave an inefficiency and other tools are needed: Pigouvian taxes, quotas, tradable permits, or mergers that internalize the effect (Varian §35.3 and §35.6).

Interpretation. The question is a compressed statement of the Coase theorem and its limits. Under its hypotheses, assigning rights decides only the transfers; that is why M&R turn the theorem into a research programme: look for the transaction costs that stop the parties from bargaining to the efficient y, and explain organizations as ways to reduce them (p. 38). "Property rights solve externalities" is the benchmark, not the conclusion.

Verification. With $a = \ln 1.5$: $e^{-a} = 2/3$, so WTP $= 100/3 \approx 33.33$; $e^{a} - 1 = 0.5$, so WTA $= 50$; and $33.33 < 40 < 50$, so the efficient outcome flips with the assignment. Utility checks: $\ln(100 - 100/3) + \ln 1.5 = \ln 100$ and $\ln(150) = \ln 100 + \ln 1.5$ (`check_C.py`).

Answer. True only in the Coasean benchmark (well-defined, enforceable rights; few parties; costless bargaining with full information): rights turn the externality into a tradable good and trade reaches an efficient level. Even then the assignment changes the distribution, and with wealth effects it changes the efficient level itself (WTP 33.33 vs. WTA 50 around a gain of 40). With many parties, private information or costly enforcement, rights alone do not remove the inefficiency.

8 Varian Ch. 35, Review Question 2

Foundations · block B1

Source: Varian Ch. 35, Review Question 2, page 687 (statement in the book; not reproduced here).

What is asked. Decide whether quasilinear preferences remove the distributional consequences of how property rights over an externality are assigned.

Method. Quasilinear preferences and the Coase theorem (Varian §35.2, pp. 670–672, Figure 35.2); M&R's value maximization principle and Coase theorem (pp. 36–38), of which Varian's result is the two-person special case; symmetric Nash bargaining to fix a split.

Solution. Verdict: false. Quasilinearity removes the effect of the assignment on the *efficient level* of the externality, not on the *distribution*. Varian's answer (p. A29) is a bare "False"; the argument follows.

(a) The level is independent of the assignment. Let $u_A = m_A + v_A(s)$ and $u_B = m_B + v_B(s)$, where s is the amount of smoke and m_i money, with $m_A + m_B = \bar{M}$. Pareto efficient allocations solve

$$\max_{s,\, m_A,\, m_B} \; m_A + v_A(s) \quad \text{s.t.} \quad m_B + v_B(s) \geq \bar{u}_B, \;\; m_A + m_B = \bar{M}.$$

The constraint binds, so $m_A = \bar{M} - \bar{u}_B + v_B(s)$ and the problem becomes $\max_s \; v_A(s) + v_B(s) + \text{const}$. The first-order condition

$$v_A'(s^*) \;=\; -\,v_B'(s^*)$$

does not involve $\bar{u}_B$, $\bar{M}$ or the endowment. Every interior Pareto efficient allocation has the same s^*; the Pareto set is a horizontal line in the smoke–money box (Varian Figure 35.2), and only the money holdings vary along it. This is the Coase theorem in Varian's form: "the efficient amount of the good involved in the externality is independent of the distribution of property rights" (p. 671). It needs both parties to hold enough money for the compensating payments, so that the solution is interior (M&R's condition (iii), p. 35).

(b) The distribution is not. The assignment fixes the disagreement point, and the bargaining gains are divided starting from it.

Example. $v_A(s) = 6s - s^2$ (the smoker's benefit, maximal at $s = 3$) and $v_B(s) = -s^2$ (the nonsmoker's harm). Joint value $v_A + v_B = 6s - 2s^2$ is maximal at $s^* = 1.5$, where it equals 4.5; it is 0 both at $s = 0$ and at $s = 3$.

	A holds the right to smoke	B holds the right to clean air
Without trade	$s = 3$: A gets 9, B gets -9	$s = 0$: both get 0
Deal	B pays A an amount T to cut to $s = 1.5$	A pays B an amount T to smoke $s = 1.5$
Acceptable T	A: $6.75 + T \geq 9$; B: $-2.25 - T \geq -9$ $\Rightarrow T \in [2.25, 6.75]$	A: $6.75 - T \geq 0$; B: $T - 2.25 \geq 0$ $\Rightarrow T \in [2.25, 6.75]$
Nash split of the gain 4.5	$T = 4.5$ from B to A	$T = 4.5$ from A to B
Final payoffs (net of initial money)	$u_A = 11.25$, $u_B = -6.75$	$u_A = 2.25$, $u_B = 2.25$

The smoke level is 1.5 and total surplus is 4.5 under both assignments. But A is better off by 9 when he holds the right and B is worse off by 9: the right is an asset worth $v_A(3) - v_A(0) = 9$ to whoever starts with it, and the direction of the payment flips. Within $[2.25, 6.75]$ the size of the payment depends on bargaining power, which quasilinearity also leaves open.

Interpretation. Quasilinear utility is the two-person version of M&R's "no wealth effects" (item 27 derives it). Then the value maximization principle separates *what* is done (efficiency alone chooses s^*) from *who gets what* (the assignment and bargaining power choose the transfers, M&R p. 38). Coase's result is a statement about efficiency, not fairness: choosing who holds the right remains a distributive decision.

Verification. $v_A'(1.5) = 6 - 3 = 3 = -v_B'(1.5)$. $v_A(1.5) = 6.75$, $v_B(1.5) = -2.25$, $v_A(3) = 9$. Payoffs add up: $11.25 - 6.75 = 4.5 = 2.25 + 2.25$ (`check_C.py`).

Answer. False. With quasilinear preferences every efficient allocation has the same level of the externality ($v_A'(s^*) = -v_B'(s^*)$; in the example $s^* = 1.5$ under either assignment), so the assignment of rights does not affect efficiency. It still decides who pays whom: payoffs $(11.25, -6.75)$ when the smoker holds the right versus $(2.25, 2.25)$ when the nonsmoker does.

9 Cabral 5.1 Front yards

Foundations · block B1

Source: Cabral 5.1 Front yards, page 117 (statement in the book; not reproduced here).

What is asked. Judge whether the positive externality from a well-tended front yard is important, and whether the market outcome is inefficient.

Method. Externalities and market failure, Pigou corrections, social norms and the Coase theorem (Cabral §5.1, pp. 93–99); the Coase theorem and its failure conditions (M&R Ch. 2, pp. 35–39; notes, Aula 2, "The Coase theorem"). A two-neighbour model makes the comparison exact.

Solution.

(a) A two-sided externality: the model. Two neighbours $i = 1, 2$ choose yard care $x_i \geq 0$. Care gives the owner a benefit ax_i (enjoyment, resale value), gives the neighbour a spillover sx_i with $s > 0$, and costs $x_i^2/2$. Neighbour 1's payoff is

$$u_1 = ax_1 + sx_2 - \tfrac{1}{2}x_1^2,$$

and symmetrically for neighbour 2. The externality runs both ways: each is a producer and a receiver of it.

Without bargaining (each chooses alone), $\partial u_i/\partial x_i = a - x_i = 0$, so $x_i^N = a$. The spillover s does not enter the choice.

Efficient care maximizes $u_1 + u_2$: $\partial(u_1 + u_2)/\partial x_i = a + s - x_i = 0$, so $x_i^* = a + s > a$. Each neighbour's gain from moving both to x^* is

$$u_1(x^*) - u_1(x^N) = \left[(a+s)^2 - \tfrac{1}{2}(a+s)^2\right] - \left[a^2 + sa - \tfrac{1}{2}a^2\right] = \tfrac{1}{2}s^2,$$

so the joint loss of the uncoordinated outcome is s^2. With $a = 1$, $s = \frac{1}{2}$: care 1 instead of 1.5, and the pair loses $\frac{1}{4}$. Without some mechanism, yards are under-tended: the market outcome is inefficient, as with any positive externality (p. 94).

(b) Is the externality important? Per household the stakes are small: s^2 is second order in the spillover, and a neighbour's yard is a modest part of the value of one's own home. Two forces make it larger than it looks. A street has many neighbours, and the effects add up across all who see the yard (the externality is close to a local public good, p. 95); and the effect is capitalized into house prices, so a run-down block lowers every owner's wealth. Even so, compared with pollution or congestion (pp. 94–95), it is a minor externality.

(c) Can the neighbours bargain to the efficient level (Coase)? With two neighbours the Coase conditions nearly hold: few parties, a visible activity, low bargaining costs, and no large wealth effects. The two-sided structure helps further: the deal is reciprocal ("I keep my yard if you keep yours"), so with symmetric neighbours no money needs to change hands; each gains $\frac{1}{2}s^2$. Two problems remain.

(i) *Enforcement.* Given that the neighbour keeps $x_2 = a + s$, neighbour 1's best reply is still $x_1 = a$, which gains him $\frac{1}{2}s^2$ over keeping the deal. The agreement needs enforcement. Neighbours interact for years, so a social norm with the threat of withdrawn goodwill can sustain it (the repeated-game argument of p. 96).

(ii) *Many parties and measurement.* Along a whole street, each owner free rides on the others (the free-riding logic of p. 94), and "well tended" is hard to specify and verify; private information about how much each values tidiness makes a full agreement harder still (Aula 2, failure conditions of Coase).

Where these costs bite, communities use other instruments: homeowners' associations with covenants (a contract signed when the house is bought), municipal ordinances (direct regulation, p. 96) and the social norm itself. A Pigou subsidy (p. 96) is possible in principle but impractical because yard care is hard to measure.

Interpretation. This is the Coase theorem at small scale. The externality creates an inefficiency only to the extent that transaction costs block bargaining; in a two-household setting those costs are low and norms do most of the work, so the residual inefficiency is small. On a large street, free riding and measurement make explicit rules (covenants) the efficient organizational response, which is M&R's corollary that efficient organization minimizes transaction costs (p. 38).

Verification. The sympy script `check_B.py` confirms $x^N = a$, $x^* = a + s$, the gain $\frac{1}{2}s^2$ per neighbour, the deviation gain $\frac{1}{2}s^2$ against the reciprocal deal, and the loss $\frac{1}{4}$ at $a = 1$, $s = \frac{1}{2}$. Limit case: $s \to 0$ gives $x^* \to x^N$ and no loss; the inefficiency rises with the square of the spillover.

Answer. Yes, the outcome without bargaining is inefficient: each owner sets $x^N = a < x^* = a + s$ because the neighbour's benefit is ignored, losing $\frac{1}{2}s^2$ per household. The externality is modest per house but adds up across a street and is capitalized into prices. Two neighbours can largely fix it by a reciprocal, norm-enforced deal (Coase); on a whole street free riding and measurement push the solution towards covenants or ordinances.

10 Cabral 7.6 Hernan Cortez

Foundations · block B2

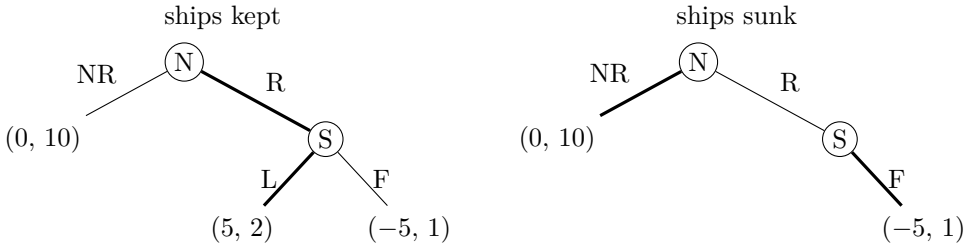
Source: Cabral 7.6 Hernan Cortez, page 180 (statement in the book; not reproduced here).

What is asked. Explain the strategic value of Cortés sinking his own ships on arrival in Mexico, given that his force expected resistance from the native population.

Method. Sequential games, backward induction and the value of credible commitment (Cabral §7.2, pp. 167–171, Figures 7.7–7.9: a player gains by removing or penalizing its own options). Course link: M&R on achieving commitment, where William the Conqueror and Cortés are the examples (Ch. 5, pp. 133, 139; notes, Aula 2, "Commitment, reputation and the ultimatum game").

Solution.

(a) The game without commitment. Model the Cabral entry game of Figure 7.7 with the roles adapted. The natives (N) move first: resist (R) or not (NR). If they resist, the Spaniards (S) choose to fight (F) or leave by sea (L). Illustrative payoffs (N, S), chosen only to respect the ordering of the story:



Not resisting hands the Spaniards the full prize (10); a resisted campaign that ends in withdrawal leaves them with little (2) and the natives better off (5); a fought war is costly for both (−5, 1). Backward induction (thick branches): if resisted, the Spaniards compare 2 (leave) with 1 (fight) and leave. The natives anticipate this and compare 5 (resist) with 0 (not resist), so they resist. Outcome: $(5, 2)$. The Spaniards' threat to fight to the end is not credible, exactly like the incumbent's threat in Figure 7.7 (p. 169).

(b) Sinking the ships. Destroying the fleet deletes the branch L. Now a resisted Spaniard can only fight, so resisting gives the natives $-5 < 0$ and they do not resist. Outcome: $(0, 10)$. The value of commitment is $10 - 2 = 8$ for the Spaniards, and the move is worth making whenever

the ships' value is below 8. This is Cabral's bond of Figure 7.8 taken to the limit: instead of a penalty for retreating, retreat becomes impossible (equivalently, the Spaniards now move first, as in Figure 7.9, p. 171).

(c) Two audiences. The quotation itself ("I could act more securely") stresses a second effect. Before the ships were sunk, any of Cortés's own men could press to go back; afterwards no one could, so each soldier's best response changed from hedging to fighting hard. The same commitment therefore works on the rival (it changes what the natives expect the Spaniards to do) and inside the organization (it removes an exit option, aligning the members' interests with the leader's plan).

(d) Conditions and costs. Commitment has value only if (i) it is irreversible: a fleet that can be refloated commits nothing; (ii) the other side observes it, since a commitment no one knows about cannot change their choice; and (iii) the cost is accepted: M&R note that such noncontractual commitments "may be worthwhile, but costly", since the fleets were perfectly good (p. 139). If the natives resist anyway (they did not see the act, or misread it), the Spaniards are stuck with 1 instead of the 2 they could have had by leaving.

Interpretation. Bargaining power comes from what a player can credibly do next, and a player can raise its power by *reducing* its own options. The same logic runs through the course: the proposer in an ultimatum game is committed to walking away, specialized investments commit a firm to a strategy (Apple's Macintosh plant, M&R p. 133), and the hold-up problem of item 30 is the reverse case, where a sunk investment weakens the investor's position.

Verification. The script `check_B.py` solves both trees by backward induction: $(5, 2)$ with the ships, $(0, 10)$ without, a commitment value of 8. Robustness: the result needs only the orderings $u_S(\text{leave}) > u_S(\text{fight})$, $u_N(\text{resist, leave}) > u_N(\text{no resistance}) > u_N(\text{resist, fight})$ and $u_S(\text{no resistance}) > u_S(\text{resist, leave})$; any payoffs with these orderings give the same two equilibria.

Answer. Sinking the ships was a credible commitment. With the fleet available, a resisted Spaniard would rather leave than fight, so resistance pays and the natives resist. Without it, the only option is to fight, which makes resistance unprofitable and (in the model) deters it; it also stops Cortés's own men from hedging. The move is valuable only because it is irreversible, observed and costly, as in Cabral §7.2 (pp. 169–171).

Tier 2 — Exam level

11 M&R Food for Thought 1.3

Exam · block A1

Source: M&R Food for Thought 1.3, page 18 (statement in the book; not reproduced here).

What is asked. Given the 1991 Treasury-auction scandal at Salomon Brothers, judge whether the episode shows that the firm's incentive scheme, as described in Chapter 1, was fundamentally flawed.

Method. The Salomon case (M&R Ch. 1, pp. 9–12): bonus close to an individual piece rate on attributed profit (pp. 10–11), its known side effects (p. 11), and the May 1990 reform paying part

of the bonus in stock held in trust for five years (pp. 11–12). The patterns of pp. 16–17: authority where the information is, incentives and delegation as complements. Complementarity and coherent systems (Ch. 4, pp. 108–113). The question requires separating three levers: *incentives* (what is rewarded), *selection* (who is attracted and promoted) and *control* (what actions are allowed, monitored and punished).

Solution.

Incentives: a real defect, but not the one the reform targeted. The bonus rewarded measured profit attributed to a desk (p. 10). Two costs were missing from that measure: legal and regulatory risk, and damage to the firm's reputation, a firm-wide asset. A trader who corners an auction collects the measured profit; the firm bears the legal penalty, the lost customers and the fall in share price. When the downside is mostly the firm's and the upside largely the trader's, high-powered pay on measured profit rewards excessive risk, including the risk of breaking rules. In that precise sense the scheme was flawed: it paid for a measure that omitted important costs. M&R had already listed related side effects — little cooperation, focus on own accounts, "stealing" other departments' profits (p. 11).

The 1990 reform aimed at this problem: stock held in trust for five years ties each bonus to the firm's long-run value (p. 11), and the scandal did cut that value sharply. But stock pay is a *group* incentive. One managing director owns a tiny fraction of the firm, so his share of the firm-wide loss is negligible next to the personal gain from a large trading profit. Stock ownership can encourage cooperation and a long-run view across the whole staff; it cannot deter one individual's rule-breaking. The reform was also barely in place when the violations began (February 1991). The episode therefore does not refute the reform's logic, but it shows its limits.

Selection and culture. Profits were "the sole source of power, prestige, status, and respect" (p. 10), and the pay system attracted individualistic, competitive risk-takers (p. 10). One of the theories offered for the violation — a competitive need to beat the regulators rather than greed — is a selection explanation: the marginal incentive may matter less than *who* the system attracts and promotes. M&R expected the reform to change the type of people Salomon attracted only "in the long run" (p. 11).

Control. Senior executives learned of the violations and did not act. This is a failure of monitoring and governance, not of the trader's pay formula. Salomon gave traders enormous discretion because they held the information (p. 17), and the book stresses that such delegation needs incentives to go with it. The case adds the other half: discretion over actions with catastrophic, firm-wide downside also needs limits, compliance and reporting that someone is paid — and willing — to enforce. The executives' own incentives pointed the other way: reporting promptly would have meant immediate costs for them.

Verdict. Not fundamentally flawed as a device to elicit effort and use traders' local information — that was the scheme's purpose and it achieved it (p. 11). Flawed as a *system*: profit-based pay, a culture that ranked people by profit, selection of aggressive risk-takers and weak controls were mutually reinforcing, a coherent pattern in the sense of pp. 108–113. Because the parts are complements, changing one of them (adding stock to the bonus) could not move the whole system; the fix had to reach controls, leadership and culture together.

Interpretation. Judging an incentive scheme by one scandal confuses three levers. The right question is which lever failed. Here all three contributed, and the case is a warning against reading

pay alone as "the incentive system": incentives, selection and control form one design.

Verification. Counterfactual tests. With the same bonus but strong controls (position limits, compliance checks, senior managers who report violations), the violation would have been far less likely: so pay alone is not sufficient to explain it. With flat salaries, the 1980s trading profits and the use of traders' information would probably have been lost (pp. 10–11): so removing incentive pay is not the fix. The market evidence points the same way: the reform raised the share price by 7–10% (p. 12), so investors valued its incentive logic.

Answer. No, not fundamentally: the scheme did what it was designed to do, eliciting effort and using traders' information. But it was flawed in what it measured (attributed profit that ignored legal and reputational risk), and the 1990 stock reform is a group incentive that cannot deter one individual. The scandal mainly exposes failures of selection, culture and control, which formed a coherent system with the pay scheme; fixing it required changing all of them together.

12 Cabral 3.14 Intel

Exam · block A2

Source: Cabral 3.14 Intel, page 63 (statement in the book; not reproduced here).

What is asked. Explain Intel's sustained competitive advantage over its rivals and weigh the explanatory power of the four causes of §3.5: impediments to imitation, causal ambiguity, strategy and history.

Method. Sources of sustained competitive advantage (Cabral §3.5, pp. 56–59): impediments to imitation (legal and other, p. 57), causal ambiguity and tacit knowledge (p. 57), strategy (p. 57), history and the learning curve (pp. 58–59). Facts about Intel are restricted to what Cabral reports: leadership in quality and efficient production, very high fab set-up costs and a much smaller rival AMD (§5.3, pp. 103–104); exclusivity rebates and the resulting antitrust cases (§12.2, pp. 315–316; §12.4, pp. 321–322); reliance on production efficiency and the learning curve rather than on intellectual property (§15.3, p. 386).

Solution.

(a) Impediments to imitation. Two barriers apply. The first is scale: the set-up cost of a fab is so high that a rival can hardly match Intel's size (p. 103), so unit costs stay lower for the leader even when the technology is known. This is less "imitation is impossible" than "imitation at the same scale is unprofitable". The second is legal protection: patents and secrecy exist, but Cabral notes that Intel relies primarily on production efficiency and that its success "is more a case of execution than building valuable IP assets" (p. 386). Unlike Xerox's plain-paper patent (p. 57), law is not the main barrier. *Weight: high for scale, low for patents.*

(b) Causal ambiguity. Producing chips efficiently means running a fab with high yields at each new generation, a body of process know-how built by experience and held by teams as tacit knowledge (p. 57). A rival that hires some of Intel's engineers does not acquire the system, much as Toyota's procurement system could not be copied piece by piece (p. 57). *Weight: moderate to high*, because it explains why the efficiency lead persists rather than spreading.

(c) Strategy. Cabral documents an aggressive strategy: all-unit discounts that made buying any AMD chips expensive for PC makers, an instance of raising rivals' costs (§12.2, pp. 315–316), which led to large antitrust settlements and fines (§12.4, pp. 321–322); he suggests aggressive pricing is one reason AMD did not grow into a closer competitor (p. 103). Moving quickly from one generation to the next (p. 386) is also a strategic choice: it keeps the leader ahead on the learning curve of each new product. *Weight: high*, though part of it (the rebates) was later restricted by antitrust.

(d) History. The learning-curve mechanism converts an early lead into a lasting one: the firm that has produced more has lower costs, which lets it price lower and produce more again, as in Boeing's case (pp. 58–59). An early dominant position in microprocessors (Cabral lists Intel as the dominant firm since the 1980s, p. 104) also matters because buyers and complementary products (software, operating systems) are built around the dominant design (complementary products are the setting of Exercise 13.7). *Weight: high*, as the starting condition the other three amplify.

(e) Weighing them. The causes are complements, not rivals. History gave an early lead; the learning curve and fab scale (history plus a scale-based impediment) made the lead cumulative; tacit process know-how (causal ambiguity) kept rivals from closing the gap by copying; and strategy (pricing, rebates, fast generations) protected the share the cost advantage earned. In Cabral's own reading the core is production efficiency along the learning curve (p. 386), so the ranking is: scale and learning (history) and strategy first, causal ambiguity second, legal protection last.

Interpretation. For the course, the useful lesson is the one Aula 1 draws from Toyota: an advantage that rests on a coherent system of mutually reinforcing elements (scale, know-how, pricing, product cadence) is harder to imitate than one resting on a single asset, because a rival must copy the whole system at once (M&R Ch. 4 complementarities; Cabral p. 57). Sustained is not permanent: the same framework says the advantage erodes if the learning-curve lead is lost, as with Lockheed and McDonnell Douglas against Boeing (p. 59).

Verification. Consistency check with §3.5's evidence: performance differences within an industry are large and persistent, and only about 20% of the variance in profit rates is explained by industry and size (p. 56). A single dominant firm whose lead persists across generations is what the four causes predict, and the antitrust record (pp. 321–322) shows that at least part of the advantage was defended by conduct, not only earned by efficiency.

Answer. Intel's advantage is best explained by history and the learning curve combined with fab scale economies (an early lead that became a cost lead), protected by strategy (aggressive pricing, exclusivity rebates, fast product generations) and made hard to copy by tacit process know-how (causal ambiguity). Legal protection (patents) is the weakest of the four in Cabral's account (p. 386). The causes reinforce each other, which is why the advantage persisted.

13 Cabral 3.11 Smart car

Exam · block A2

Source: Cabral 3.11 Smart car, page 62 (statement in the book; not reproduced here).

What is asked. Using §3.4, identify the likely sources of strain in the jointly owned Daimler-Benz and Swatch venture that produced the Smart car, which ended with Daimler buying out

Swatch's share.

Method. Firm boundaries as a balance between investment incentives (specific assets, hold-up) and performance incentives; intermediate forms between market and hierarchy (Cabral §3.4, pp. 54–56). Supporting ideas: tacit knowledge and firm culture (§3.5, p. 57); integration trades hold-up for agency problems (§13.1, pp. 338–339). Course contrast: NUMMI and complementarities (M&R Ch. 4; notes, Aula 1, "NUMMI").

Solution. A joint venture is one of the intermediate forms of p. 55: neither a market contract nor a single firm. §3.4 predicts it inherits problems from both sides. The exercise gives only the ownership history; the sources below are what the theory points to, not reported facts about the partners' disputes.

(1) Specific investments and hold-up between the owners. Each partner brings assets that, once committed, are valuable mainly inside the venture: a carmaker brings automotive engineering and production, a watchmaker brings design and consumer-brand know-how. Once sunk, each contribution is a specific asset, and each partner can threaten to withhold cooperation to renegotiate the terms (the hold-up problem of p. 55). A 50/50 ownership split also means each partner reaps only half the return on any further investment it makes in the venture, so both under-invest relative to the joint optimum. Item 30 (Cabral 13.10) computes the size of this effect: with two investors and an even split, total surplus falls from $\frac{1}{2}$ to $\frac{3}{8}$.

(2) Weak performance incentives inside, weak control outside. A joint venture removes the market discipline that a supplier relationship has (neither partner can easily switch to another partner, p. 55) without creating the single line of authority of a firm. With shared control, every major decision (design, cost targets, pricing, timing) must be agreed by two owners whose objectives differ: each owner cares about the venture's profit and also about its effect on its own core business and brand. Joint ownership with divergent objectives is the agency problem of §3.3 (p. 52) with two principals.

(3) Two organizational systems that do not mix. A carmaker and a watchmaker have different capabilities, routines and cultures, much of them tacit (p. 57). A car designed around one firm's ideas and built with the other's production system requires the two systems to be merged inside a single product. The complementarity logic of Aula 1 says that a mix of practices taken from two coherent systems can perform worse than either system alone, because each practice is valuable only alongside its complements.

(4) The resolution is the theory's fix. Daimler's buyout ended joint control: one owner now made all decisions and received the whole return on investment, the integration solution to hold-up (p. 55; §13.1, p. 338). The cost, as §3.4 and §13.1 warn, is the loss of the partner's own incentive: Swatch's design and marketing input, if still used, now had to be bought or managed from inside.

(5) The contrast with NUMMI. NUMMI was also a joint venture between carmakers with different systems, but it was run on *one* system: Toyota's production system governed the plant, and GM's role was ownership and learning (notes, Aula 1). The internal strain predicted by points (2)–(3) was therefore small, since there was no merging of two systems inside the operation. NUMMI's limitation was elsewhere: GM could not transfer the system piece by piece to its own

plants, which is again the complementarity point. Smart illustrates the opposite configuration: the systems had to be merged within the venture itself, so the strain appeared inside it.

Interpretation. Joint ventures look like a way to combine the best of two firms, but §3.4 says they combine the hold-up exposure of separate ownership with the weak incentives of shared control. They work best when each partner's contribution is separable or when one partner's system clearly governs operations (NUMMI); they strain when two systems must be fused (Smart), and the usual exit is that one partner buys the other out.

Verification. Cross-check against the theory's predictions. If strain came from hold-up and shared control, then (i) the venture should end with consolidation of ownership rather than with a looser market contract, and (ii) the natural buyer is the partner in whose core business the product lies (a car lies in the carmaker's). Both match the outcome reported in the exercise (Daimler took complete control in November 1998).

Answer. §3.4 points to (1) specific investments by both owners and hold-up between them, with a 50/50 split under-rewarding each partner's investment; (2) shared control between owners with different objectives, i.e. weak incentives without a single authority; and (3) two different, largely tacit organizational systems that had to be fused in one product. Daimler's buyout is the integration fix. NUMMI, run on Toyota's single system, avoided the internal strain.

14 M&R Food for Thought 4.2

Exam · block A3

Source: M&R Food for Thought 4.2, page 119 (statement in the book; not reproduced here).

What is asked. Identify the core competencies behind Swiss dominance of mechanical watchmaking, explain how battery-powered electronic watches changed them, and say what else the new watchmakers produce.

Method. Core competencies as scale economies in development shared across products, and economies of scope (M&R Ch. 4, pp. 107–108, with the Casio example on p. 107). Complementarities, coherent systems and the shift of the best system (pp. 108–113, Figure 4.3). Innovation attributes (p. 92, pp. 112–113). Ford's coherent but superseded strategy as the parallel case (p. 109, p. 111).

Solution.

The Swiss competencies. Accuracy in a spring-driven watch is a mechanical achievement. The competencies were (i) the design of micro-mechanical movements; (ii) precision fabrication and finishing of tiny parts (springs, gears, escapements, jewelled bearings); (iii) highly skilled hand assembly and regulation, reproduced by long apprenticeship; (iv) a dense network of specialized small suppliers coordinated by long-standing relationships; and (v) a reputation for quality and a distribution channel through jewellers. These pieces formed a *coherent system* (pp. 111–112): small batches, craft labour paid for skill, quality achieved by the worker rather than by the machine, high prices justified by brand and craftsmanship. Each element raised the return to the others — complementarity in the sense of p. 108.

What the new technology changed. With a quartz oscillator driven by an integrated circuit and a battery, accuracy comes from electronics, and it is better and cheaper than the best hand-regulated movement. The relevant competencies became semiconductor design and fabrication, displays (LCD), miniature batteries, and high-volume automated assembly. These have very large scale economies, and they are *shared inputs* across many products: exactly the core competencies of p. 107. Casio's case in the text illustrates the logic: scale economies in LCDs made calculators, wristwatches and electronic address books cheaper together, and large calculator sales lowered display costs and made watch entry profitable (p. 107). The value of the Swiss competencies for the mass market fell sharply: mechanical skill no longer bought accuracy, and the craft network could not match semiconductor-scale costs.

Why the incumbents struggled to adapt. In Figure 4.3's terms (pp. 111–113), technology moved the profit landscape so that a different coherent system — mass production of electronic modules, automated assembly, low prices, wide distribution — became the best one. Managers inside the craft system can adjust *locally* (cheaper springs, better tooling), but local moves keep them on the old peak. Reaching the new one requires changing many complementary elements at once — skills, equipment, suppliers, distribution, pricing — and partial moves lose money, the "valley" of the Aula 1 notes. It also requires information no one in the organization has (innovation attributes, pp. 92, 113): watchmakers did not know semiconductors. In short, the competencies that made the old system excellent were not only useless for the new one, they were *complementary to the old system*, which made the jump harder.

The old system did not become worthless everywhere: it remained coherent for a luxury segment where a mechanical watch is valued as craftsmanship and jewellery rather than as a timekeeper. Broadly, the Swiss industry's later recovery paired a reorganized, low-cost quartz line with this repositioning of mechanical watches — two coherent systems for two segments.

Other products of the new watchmakers. Firms whose competencies are in electronics apply them across product lines: calculators, electronic organizers and address books (p. 107), and more generally consumer electronics, displays and the semiconductor components themselves. For them the watch is one more product that shares components and development capabilities with the rest — an economy of scope, not a craft.

Interpretation. A coherent system is a strength only for the environment it was built for (p. 109: Ford's strategy was "equally coherent" for its time). New technology can make a different coherent system optimal, and complementarities turn yesterday's competencies into today's inertia. Choosing the new pattern is a central, strategic decision with innovation attributes (p. 115).

Verification. Alternative argument by comparison: the same mechanism explains Ford's loss to GM and GM's to Toyota (pp. 16, 111). In each case the incumbent's system stayed internally consistent, while the environment changed which system was best. Limit case: an innovation that improved the spring or the escapement would have *raised* the value of Swiss skills and called only for local adjustment. What made the quartz watch destructive is that it moved the source of accuracy outside the competencies the system was built on.

Answer. Swiss competencies were micro-mechanical design, precision part-making, skilled hand assembly, a network of specialized suppliers and a quality brand — a coherent craft system. The quartz-electronic watch moved accuracy into semiconductors, displays and batteries made at large scale, the core competencies of electronics firms, which also make calculators, organizers and other consumer electronics. Complementarities and missing information made the Swiss jump to the new system slow; the craft system survived mainly in the luxury segment.

15 M&R Food for Thought 4.3

Exam · block A3

Source: M&R Food for Thought 4.3, page 119 (statement in the book; not reproduced here).

What is asked. Explain why consumer-electronics leaders (Sony, then Matsushita) bought record, film and television producers around 1987–1990.

Method. Complementarity as rising marginal profitability, $\partial^2\pi/\partial x_i \partial x_j \geq 0$ (M&R Ch. 4, p. 108 and n. 8). Strong complementarity and design attributes (p. 109). Multiple coherent patterns and coordination failure (pp. 111–113). Design decisions: synchronization and the communication of design variables such as a technical standard (pp. 104–106; the HDTV standard, p. 108). Economies of scope and core competencies (p. 107), as the contrasting motive.

Solution.

The main motive: complementarity between hardware and content. Let x_h be the firm's effort in a playback format (players, recorders, televisions) and x_c the content available in that format (records, films, programmes). Consumers buy a player for what they can play on it, and content sells only to people who own players. So more content raises the marginal profit of selling hardware, and more hardware raises the marginal profit of releasing content: $\partial^2\pi/\partial x_h \partial x_c \geq 0$, the definition of p. 108. A firm doing only one side captures only part of the joint return. Owning both lets it internalize the externality: release content in its new format early, price hardware to build the installed base, and accept losses on one side for gains on the other.

Design decisions and synchronization. A new format is a design variable with both attributes of p. 91. Its success requires a predictable fit — players, discs or tapes and a content catalogue in the same standard, available at the same time — and a misfit (a format without titles at launch) is very costly, while small differences in technical quality matter much less. Like the car launch date (p. 104), the launch of a format is best coordinated by announcing the design variables (the standard, the date) to all parties. The text's own example is HDTV: Japan's push was grounded in electronics and displays, and a central standard focused the effort (p. 108). Owning a studio and a record label puts both sides of the design decision under one management.

Coordination failure and multiple coherent outcomes. With strong complementarities there are several self-reinforcing outcomes (pp. 111–113): "no content, so no players, so no content" is coherent, and so is "rich catalogue, large installed base". Independent studios, each waiting for others, can leave a new format in the low outcome or choose a rival standard. The lesson the electronics firms drew from home-video format contests — where the format with more available titles prevailed — is that hardware quality alone does not decide the outcome. Owning a content library is a way to make the jump to the good outcome rather than wait for it.

What the motive is not. It is not mainly economies of scope or shared core competencies (p. 107). Making films and records draws on creative, talent-based capabilities quite unlike precision electronics manufacturing; little is shared on the production side. The complementarity is in *demand*, not in production costs. Other motives may also have mattered (diversification, growth), but they do not explain why these particular industries were chosen.

The trade-off. Complementarity between activities does not by itself require common ownership: licensing and contracts with independent labels and studios can coordinate a launch, and many formats have succeeded that way. Ownership pays when coordination must be repeated across formats and is hard to contract for. And it carries a cost that Chapter 4 predicts: a Hollywood studio and a Japanese electronics maker are each coherent systems with very different incentives, careers and decision rights, and the parent's top management lacks information about the creative business (innovation attributes, p. 92). Imposing one firm's policies on the other breaks the internal fit of the acquired business.

Interpretation. Complementarities explain the direction of expansion: firms grow into the activities that raise the marginal return on what they already do. They also explain why such expansions are hard to manage — each business is itself a system of complementary practices, and combining two coherent systems does not make a coherent whole.

Verification. Comparative statics with complements (p. 108, n. 8): a shift that raises the return to hardware (say, a new disc format) raises the optimal level of content, and vice versa. The prediction is that acquisitions cluster around new formats — consistent with the timing, as CDs, home video and the prospect of HDTV were all live issues in 1987–1990. If hardware and content were independent ($\partial^2\pi/\partial x_h \partial x_c = 0$), the acquisitions would have no strategic logic beyond financial diversification.

Answer. Because consumer electronics and recorded content are complements in demand: content raises the return to players and players raise the return to content. Owning studios and labels lets the firm coordinate the design decision of a new format (standard, launch timing, catalogue) and escape the "no content, no players" coordination failure. The motive is complementarity, not shared production competencies; the cost is managing a business whose coherent organization is very different from the parent's.

16 M&R Mathematical Exercise 2.1

Exam · block B1

Source: M&R Mathematical Exercise 2.1, page 53 (statement in the book; not reproduced here).

What is asked. Four families share a beach and choose how much to spend on common improvements; find the efficient total expenditure given each family's value of the spending.

Method. Value maximization principle (M&R pp. 35–38): with no wealth effects an allocation is efficient if and only if it maximizes total value, here the sum of the families' values minus the cost (Q 2.3, item 26, proves this). The improvement is a public good, so the first-order condition is the Samuelson rule: the *sum* of marginal values equals marginal cost.

Solution. **Data.** The book's value functions (gross of cost, in dollars) are

$$v_1(y) = v_2(y) = 5y - \tfrac{1}{2}y^2, \qquad v_3(y) = 7y - \tfrac{1}{2}y^2, \qquad v_4(y) = 4y - y^2,$$

where $y \geq 0$ is total spending; each dollar spent costs a dollar, so $P(y) = -y$ in the notation of Q 2.3.

Total value. Each family has utility $x_n + v_n(y)$ and the contributions satisfy $\sum_n x_n = -y$, so total value is

$$W(y) \;=\; \sum_{n=1}^{4} v_n(y) - y \;=\; (5+5+7+4)\,y - \left(\tfrac{1}{2}+\tfrac{1}{2}+\tfrac{1}{2}+1\right) y^2 - y \;=\; 20\,y - \tfrac{5}{2}\,y^2.$$

First-order condition. The marginal values are $v_1' = v_2' = 5 - y$, $v_3' = 7 - y$, $v_4' = 4 - 2y$. Setting their sum equal to the marginal cost of one dollar:

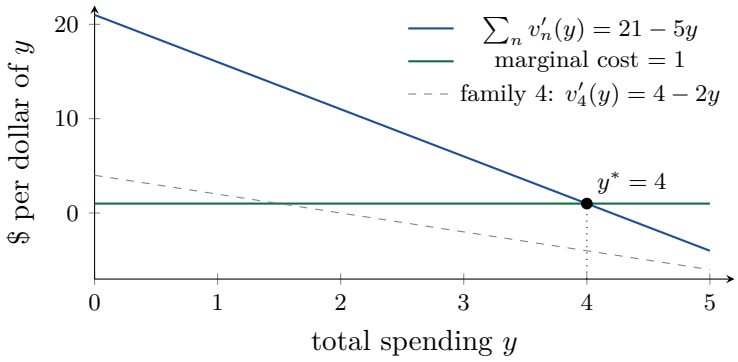
$$\sum_n v_n'(y) = 21 - 5y = 1 \quad \Longrightarrow \quad y^* = 4.$$

The second-order condition holds: $W''(y) = -5 < 0$, so W is strictly concave and $y^* = 4$ is the unique maximizer (and interior, since $W'(0) = 20 > 0$).

Values at the optimum.

Family	1	2	3	4
Gross value $v_n(4)$	12	12	20	0
Marginal value $v_n'(4)$	1	1	3	−4

Total gross value is 44, the cost is 4, so $W(4) = 40$. The marginal values sum to 1, the marginal cost.



Interpretation. The efficient project is chosen by summing the families' marginal values, not by any one family's preference: family 4 alone would stop at $y = 2$ (where $v_4' = 0$) and at $y^* = 4$ it values the improvement at zero, but the other families' high marginal values justify the extra spending. The value maximization principle says nothing yet about who pays: any contributions adding up to 4 are consistent with efficiency, and the division is a separate question (item 17 shows that a bad sharing rule can block the efficient project).

Verification. $W'(4) = 20 - 5\cdot 4 = 0$; $W(4) = 80 - 40 = 40$; $W(3.9) = 39.975$ and $W(4.1) = 39.975$, both below 40 (`check_C.py`).

Answer. Maximize $\sum_n v_n(y) - y = 20y - \frac{5}{2}y^2$: the sum of marginal values $21 - 5y$ equals the marginal cost 1 at $y^* = 4$. Efficient spending is \$4 (in the book's units), creating a total net value of 40.

17 M&R Mathematical Exercise 2.2

Exam · block B1

Source: M&R Mathematical Exercise 2.2, page 53 (statement in the book; not reproduced here).

What is asked. In the beach-improvement setting of Q 2.1, show that equal cost sharing makes one family reject the efficient project, find the largest project all families accept under equal shares, and exhibit a different spending level and cost split that every family prefers.

Method. Participation (voluntary agreement) constraints under a fixed sharing rule; the value maximization principle and its second sentence: an inefficient allocation is strictly Pareto dominated by a value-maximizing one once transfers are free (M&R p. 36; the construction in Q 2.3, item 26).

Solution. Values as in item 16: $v_1 = v_2 = 5y - \frac{1}{2}y^2$, $v_3 = 7y - \frac{1}{2}y^2$, $v_4 = 4y - y^2$; total value $W(y) = 20y - \frac{5}{2}y^2$, maximized at $y^* = 4$ with $W(4) = 40$. Each family can refuse, in which case nothing is built and it gets 0.

(a) Equal shares at the efficient level. With equal shares each family pays $y/4$, so its net value is $n_i(y) = v_i(y) - y/4$:

$$n_1(y) = n_2(y) = 4.75\,y - \tfrac{1}{2}y^2, \qquad n_3(y) = 6.75\,y - \tfrac{1}{2}y^2, \qquad n_4(y) = 3.75\,y - y^2.$$

At $y^* = 4$: $n_1 = n_2 = 19 - 8 = 11$, $n_3 = 27 - 8 = 19$, and

$$n_4(4) = 15 - 16 = -1 < 0.$$

Family 4 values the improvement at $v_4(4) = 0$ but would pay 1: it is better off with no project and will not agree. (The net values still sum to $W(4) = 11 + 11 + 19 - 1 = 40$; the problem is the division, not the size of the pie.)

(b) Largest unanimous project under equal shares. Each family accepts y iff $n_i(y) \geq 0$:

$$n_1, n_2 \geq 0 \iff y \leq 9.5, \qquad n_3 \geq 0 \iff y \leq 13.5, \qquad n_4 = y(3.75 - y) \geq 0 \iff y \leq 3.75.$$

The binding constraint is family 4's, so the largest project everyone accepts is

$$\bar{y} = 3.75,$$

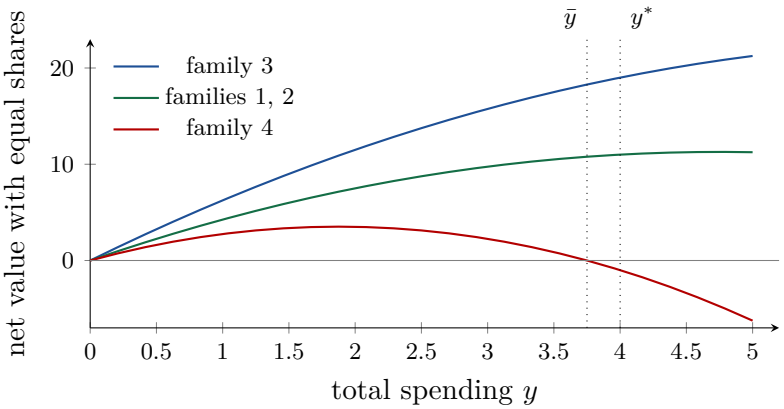
at which family 4 is just indifferent (any $y < 3.75$ it strictly prefers to nothing). Net values at $\bar{y}$: $n_1 = n_2 = 10.78125$, $n_3 = 18.28125$, $n_4 = 0$, so $W(3.75) = 39.84375$. (Family 4's *favourite* level under equal shares is 1.875, where $n_4' = 3.75 - 2y = 0$; the question asks for the largest level it would still accept.)

(c) $\bar{y}$ is inefficiently low. $\bar{y} = 3.75 < 4 = y^*$ and $W(4) - W(3.75) = 40 - 39.84375 = 0.15625 = 5/32 > 0$. Since total value is higher at $y^* = 4$, there must be a cost split at $y = 4$ that every family prefers. Following the construction of Q 2.3, give each family its net value at $\bar{y}$ plus an equal share of the gain, 5/128 each, and round to cents. Family i pays $t_i = v_i(4) - n_i(\bar{y}) - 5/128$:

Family	1	2	3	4	Total
Value $v_i(4)$	12	12	20	0	44
Contribution t_i	1.18	1.18	1.68	−0.04	4.00
Net value at $y = 4$	10.82	10.82	18.32	0.04	40.00
Net value at $\bar{y} = 3.75$, equal shares	10.78125	10.78125	18.28125	0	39.84375

The contributions cover the cost exactly ($1.18+1.18+1.68-0.04=4$), and every family is strictly better off than under the equal-share plan: the three families with high marginal values spend a little more and pay family 4 a side payment of \$0.04 to accept the larger project.

Without a payment to family 4. If contributions must be non-negative, a level just below y^* works because family 4 then values the project positively. At $y=3.9$ ($v=11.895,\ 11.895,\ 19.695,\ 0.39$), contributions $(1.10, 1.10, 1.40, 0.30)$ sum to 3.9 and give net values $(10.795,\ 10.795,\ 18.295,\ 0.09)$, again strictly above $(10.78125, 10.78125, 18.28125, 0)$.



Interpretation. A fixed cost-sharing rule ties the division (x) to the decision (y), which destroys the separation behind the value maximization principle: the low-value family's veto caps the project below the efficient level. Side payments (or any flexible cost split) restore the separation, and then unanimity and efficiency coincide. This is the organizational lesson of M&R pp. 36–38: rules that restrict transfers (equal shares, uniform prices, one-size-fits-all policies) can block value-creating decisions, so the question is always whether the transfer needed is feasible and affordable.

Verification. $n_4(3.75)=14.0625-14.0625=0$; $n_4(4)=-1$. $W(3.75)=75-35.15625=39.84375$. The contributions in the table sum to 4 and each net value exceeds its benchmark; likewise at $y=3.9$ (`check_C.py` checks all entries exactly in rationals).

Answer. With equal shares family 4's net value at $y^*=4$ is $4\cdot 4-16-1=-1<0$, so it refuses. The largest project all accept is $\bar{y}=3.75$ (family 4's $3.75y-y^2\geq 0$), with total value $39.84375<40$. Spending 4 with contributions $(1.18, 1.18, 1.68, -0.04)$ gives net values $(10.82, 10.82, 18.32, 0.04)$, strictly above $(10.78125, 10.78125, 18.28125, 0)$ for every family.

18 M&R Food for Thought 5.2

Exam · block B2

Source: M&R Food for Thought 5.2, page 164 (statement in the book; not reproduced here).

What is asked. A "signal of intent" (e.g. a price cut announcing that an incumbent will fight an expansion) is not credible if, once the rival has sunk its costs, fighting no longer pays. What means can make such signals credible?

Method. Commitment and reneging (M&R p. 133), the box on terrorism and parenting (p. 135), noncontractual commitment and its costs (p. 139), reputation (pp. 139–140), signaling (pp. 154–156), and the logic of subgame perfection: a threat counts only if carrying it out is optimal in the

subgame where it must be carried out. Commitment by removing options is also in Cabral's §7.2 and Varian's Ch. 30 ("Games of Commitment").

Solution. The problem in one inequality. Let the incumbent earn M if the rival stays out, and after entry F if it fights and A if it accommodates, with $M > A > F$. Fighting is then not optimal after entry, so the rival, reasoning backwards, ignores any announcement. A price cut today is "cheap" in the relevant sense: it is reversible and leaves $F < A$ untouched. A signal becomes credible only if it (i) changes the post-entry payoffs so that $F \geq A$, (ii) removes the option to accommodate, (iii) links this market to others or to the future, or (iv) reveals information that the incumbent really is a fighter. Each route has a cost, and the device is worth buying only if the deterrence gain $M - A$ exceeds it.

1. Sunk, specific investments. Build capacity, distribution or a plant usable only in this market. Sunk capacity lowers the marginal cost of producing a lot, so aggressive output becomes the best reply after entry. Apple's deliberately overspecialized Macintosh plant, heavily publicized, made dropping the product costly and so committed Apple to the market in the view of its rivals, employees and customers (p. 133). *Cost:* the investment itself, a plant "that may become obsolete sooner" (p. 139), and the loss if entry happens anyway.

2. Burning bridges. Destroy the option to retreat: William the Conqueror's and Cortés's fleets (p. 133), or a public, irreversible exit from the alternatives. *Cost:* the forgone option has value in the states where retreat would have been right; the fleets were "perfectly good" (p. 139).

3. Contracts with third parties. Write the threat into obligations to someone else, so that accommodating triggers a penalty. Examples: price-matching or most-favoured-customer clauses that make a price war automatic and price increases costly; supply commitments; bonds or public pledges with enforceable penalties; licensing a third producer so that a promised price cannot be raised later (the logic of M&R's Analytical Problem 5.6, p. 165). *Weakness:* renegotiation. If the third party gains by releasing the incumbent ex post, it will (pp. 134–135), so the contract must give the third party a reason to insist. *Cost:* lost flexibility and, sometimes, antitrust exposure.

4. Delegation to an agent with different preferences. Put the post-entry decision in the hands of someone who *wants* to fight: a manager known to be aggressive, or a division paid on market share or sales rather than profit. The agent's best reply after entry is to fight, which is what deters. *Cost:* the agent also fights when it should not and distorts other decisions; this is an agency cost paid in every state.

5. Reputation. If the incumbent faces many potential entrants, markets or periods, accommodating once invites further challenges. Concern for future effects "can stiffen resolve" (box, p. 135): fighting today protects profits elsewhere, so the comparison becomes F plus the value of reputation against A. Its strength grows with the frequency of similar encounters, the horizon and the profitability at stake (p. 139). *Cost:* the incumbent must actually fight sometimes, and the device weakens near the end of a known horizon.

6. Costly signals that reveal type. If the doubt concerns the incumbent's costs or toughness (private information), a deep price cut that only a low-cost or committed firm can afford separates types, as in Spence signaling (pp. 154–156). Credibility comes from the signal being too expensive for the weak type to mimic. *Cost:* profits sacrificed to signaling, which is wasteful from the joint point of view.

Common thread. All six turn talk into a changed game: they alter payoffs, options or information so that the announced action is subgame perfect. Pure announcements, however emphatic, do not.

Interpretation. Credibility is a property of incentives after the fact, not of words before it. For managers this means two things: commitment has a price, so the value of deterrence must be

weighed against lost flexibility; and devices that can be renegotiated or mimicked are weak. The same reasoning explains why delegation and organizational design, such as who decides and on what they are paid, are strategic choices and not only internal ones.

Verification. Consistency checks: each device makes the announced action optimal in the post-entry subgame (backward induction), which is the test used for the hostage and parenting examples on p. 135 and for Cortés in Cabral's Exercise 7.6 (item 10). Every source citation was checked against M&R's text (pp. 133–140, 154–156, 165).

Answer. A signal of intent is credible only if the signaled action is optimal when the time comes. The means are: sunk, market-specific investments (capacity, a specialized plant); burning bridges; contracts with third parties that penalize backing down and resist renegotiation; delegation to an agent with more aggressive preferences or incentives; reputation across markets or periods; and costly signals that a weak type cannot afford. Each costs flexibility or resources and is worth using only if deterrence gains exceed that cost.

19 Cabral 7.14 Centipede

Exam · block B2

Source: Cabral 7.14 Centipede, page 182 (statement in the book; not reproduced here).

What is asked. For the centipede game of Cabral's Figure 7.13, show by backward induction that rational players stop (d) at every node, giving payoffs $(2, 0)$, then judge whether this equilibrium is reasonable and state the rationality assumptions behind it.

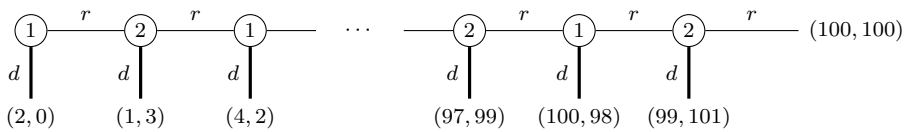
Method. Backward induction and subgame perfection (Cabral §7.2, pp. 167–169); the same unravelling as the finitely repeated game (Exercise 7.13, §7.3). The game is due to Rosenthal (1981) (Cabral, ch. 7 note 6). Course link: backward induction in alternating offers (notes, Aula 2) and reputation as a substitute for contracts (M&R pp. 139–140).

Solution.

(a) The game. Reading Figure 7.13 (p. 182): players alternate, Player 1 at the odd nodes and Player 2 at the even nodes; at each node the mover either stops (d) or continues (r). The stopping payoffs (Player 1, Player 2) are

$$\text{node } 2j-1 \text{ (P1)}: \ (2j,\ 2j-2), \qquad \text{node } 2j \text{ (P2)}: \ (2j-1,\ 2j+1), \qquad j = 1, \ldots, 50,$$

which reproduces every vector printed in the figure: $(2, 0)$, $(1, 3)$, $(4, 2)$, $(3, 5)$, $(6, 4)$ at the start and $(97, 99)$, $(100, 98)$, $(99, 101)$ at the end, with $(100, 100)$ if Player 2 continues at the last node. The pattern implies 100 decision nodes; the argument below does not depend on the length.



(b) Backward induction. Start at the last node and move left; at each node the mover compares stopping with the continuation value found one step earlier.

Node	Mover	Payoff to mover if d	Payoff to mover if r	Choice	Value carried left
100	P2	101	100 (end)	d	$(99, 101)$
99	P1	100	99 (P2 stops at 100)	d	$(100, 98)$
98	P2	99	98 (P1 stops at 99)	d	$(97, 99)$
$\vdots$				$\vdots$	
$2j$	P2	$2j+1$	$2j$	d	$(2j-1, 2j+1)$
$2j-1$	P1	$2j$	$2j-1$	d	$(2j, 2j-2)$
$\vdots$				$\vdots$	
2	P2	3	2	d	$(1, 3)$
1	P1	2	1	d	$(2, 0)$

The general step: at a P1 node $2j-1$, continuing leads (by the step already solved) to P2 stopping at $2j$, which gives P1 $2j-1 < 2j$; at a P2 node $2j$, continuing leads to P1 stopping at $2j+1$, which gives P2 $2j < 2j+1$. At every node the mover gains exactly one unit by stopping before the other does. Hence d at every node is the unique subgame-perfect equilibrium (preferences are strict), and the game ends at once with $(2, 0)$.

(c) Is it reasonable? Normatively it is compelling step by step, yet the outcome is very poor: total payoff 2 against 200 if both always continued, and *both* players would get at least 97 by reaching the last nodes. Three objections:

(i) *Evidence.* In laboratory versions of the game, few subjects stop at the first node; most play continues for several rounds before someone stops (McKelvey and Palfrey, 1992, *Econometrica* 60(4)). Cabral's Exercise 7.16 invites exactly this kind of test.

(ii) *Off-path reasoning.* The argument asks Player 2 at node 2 to assume Player 1 will stop at node 3, although reaching node 2 is evidence that Player 1 does *not* play as the theory says. Once one deviation has been observed, it is not obvious why the rest of the theory should be believed.

(iii) *A small doubt is enough.* If Player 2 assigns even a small probability to Player 1 being a "cooperator" who continues, a rational Player 1 gains by imitating that type for a while, and continuation for many rounds becomes equilibrium behaviour (the reputation logic of Kreps, Milgrom, Roberts and Wilson, 1982; M&R pp. 139–140).

(d) The rationality assumptions. The prediction needs (1) each player is rational (maximizes own payoff, with no altruism or fairness concerns, cf. M&R FFT 5.1, item 20); (2) payoffs and the game are common knowledge; and (3) *common knowledge of rationality* to depth about 100: P1 must know that P2 knows that P1 knows ... that the last mover is rational, and must expect this belief to survive after deviations. Assumption (3) is the demanding one, and it is the one the experiments and the reputation argument relax.

Interpretation. The centipede is a model of trust in a relationship with a known end. Each period both gain by continuing, but each is tempted to grab the slightly larger share now; backward induction from the known last date unravels cooperation completely, as in the finitely repeated prisoner's dilemma (Exercise 7.13). For organizations the lesson is that long relationships sustain cooperation through reputation and uncertainty about the other side's type, not through the logic

of a fixed, commonly known horizon; and the same backward-induction tool that fails here as a prediction is the right tool for alternating-offers bargaining with few rounds.

Verification. (1) The script `check_B.py` runs backward induction over all 100 nodes, recovers d everywhere and $(2, 0)$, and checks that every stopping vector printed in the figure matches the formula. (2) Stronger result: every Nash equilibrium (not only the subgame-perfect one) ends at node 1. Suppose an equilibrium path reached some node $k > 1$ and ended there with a stop by the mover at k (or at the terminal vector after node 100). The mover at $k - 1$ would then get one unit less than by stopping at $k - 1$ (the computation in (b)), and at the terminal vector P2 would gain 1 by stopping at node 100; so the profile is not a Nash equilibrium. Only paths that stop at node 1 survive.

Answer. Backward induction: at node $2j$ Player 2 gets $2j+1$ by stopping and $2j$ by continuing; at node $2j - 1$ Player 1 gets $2j$ against $2j - 1$. So d is chosen at every node and the unique SPE gives $(2, 0)$, against up to $(100, 100)$. Every Nash equilibrium has the same outcome. The prediction relies on rational, self-interested players and common knowledge of rationality to depth about 100. It is unreasonable as a forecast: experiments show prolonged continuation, and a small doubt about the other's type rationalizes it.

20 M&R Food for Thought 5.1

Exam · block B2

Source: M&R Food for Thought 5.1, page 163–164 (statement in the book; not reproduced here).

What is asked. Suppose people are partly "hard-wired" with a taste for vengeance and a conscience, a preference for honesty and fairness (Frank's evolutionary argument). How would such preferences change the chapter's analyses of commitment and bargaining?

Method. Commitment and reneging, and the grab game of Table 5.1 (M&R pp. 133, 138–139); reputation (pp. 139–140); bargaining with private information and the honest-reporting constraints of Tables 5.2–5.3 (pp. 140–143); the ultimatum game and its experimental evidence (Varian 2014, pp. 582–583). Formally, the preferences add a psychic term to material payoffs, and we ask which constraints of the chapter it relaxes or tightens.

Solution. 1. Emotions are commitment devices. The chapter's recurring problem is that a selfish player who keeps an option after the fact cannot promise or threaten credibly. A conscience makes *promises* credible. In the grab game of Table 5.1 (p. 138) a selfish firm grabs whatever the other does ($3 > 2$ and $-1 > -2$). If grabbing costs a guilt payoff g, refraining becomes dominant when $g > 1$, the mutual-refrain outcome is an equilibrium, and the specific investment that opportunism "completely destroys" is undertaken. A taste for vengeance makes *threats* credible. Punishing a cheater is costly ex post, which is why parents and governments facing hostage-takers are tempted to forgive (box, p. 135); a player who *enjoys* retaliating no longer faces that temptation, so the threat deters, and in equilibrium it seldom has to be carried out. Neither device burns a fleet or builds an overspecialized plant (p. 139): the commitment is carried in the person.

2. Bargaining predictions change. With selfish players a one-shot offer is pure commitment: the responder accepts anything positive and the proposer keeps almost all ("something like 99 cents", Varian p. 582). A responder known to reject offers below a threshold k receives k; vengeance is exactly the commitment to walk away that the selfish responder lacks. This is what experiments

show: average offers of about 45 cents, usually accepted, with small offers rejected at a cost to the responder (Varian pp. 582–583). Proposers then behave nearly optimally given the rejection rates, so selfishness plus correct expectations explains proposers, and fairness explains responders. In alternating offers the same logic raises each player's effective threat point, so the predicted split moves toward the norm (often 50–50), not toward the more patient player. A further consequence: when people care about *how* surplus is divided, the value maximization principle loses its clean separation between the value-creating decision and the transfers (M&R Ch. 2), because the transfer itself enters utility.

3. Honesty relaxes the private-information constraints. The impossibility on p. 142 comes from incentive-compatibility constraints that assume lying is free. If misreporting costs ψ, the seller's constraint becomes $p \geq 1.75 - 1.25\psi$ and the buyer's $p \leq 1.25 + 1.25\psi$: a conscience worth $\psi \geq \$0.20$ restores efficient trade at $p = \$1.50$. Rents paid for honesty can be partly replaced by honesty itself.

4. But the same preferences also create costs. (i) *Disagreement.* When the threshold k is private, a proposer must trade a bigger offer against a higher acceptance probability, so some offers are rejected and surplus is destroyed. This is the inefficiency of Tables 5.2–5.3 with fairness types in place of values. (ii) *Self-serving fairness.* M&R warn that "what is honest and fair is rarely as obvious in the rich settings of reality" and that people tend to see as fair what serves their own interests (pp. 138–139). Two vengeful parties with different fairness standards become a strike or a feud, not a deal. (iii) *Credibility of the preference itself.* Frank's argument works only if the traits are recognizable; otherwise the selfish can mimic them, and the commitment value collapses back into a reputation problem (pp. 139–140). Hence M&R's caution: it is "too risky to rely on others to act consistently contrary to their own selfish interests unless there is something that commits them".

Summary. Such preferences enlarge what can be committed to (more investment, less hold-up, honest reporting, offers near equal division) but add a new source of failure, disagreement driven by conflicting or privately known notions of fairness.

Interpretation. For the course: commitment does not need to be contractual, and emotions, norms and corporate culture can carry it. An organization designer should treat them as a complement to explicit incentives, not a substitute, because their strength is private information and their content is self-serving. The ultimatum evidence is also a reminder that equilibrium predictions depend on the payoffs actually in players' heads.

Verification. The arithmetic is checked in `check_D.py`: in Table 5.1, refraining is dominant iff $g > 1$; with a lying cost ψ, M&R's bounds become $1.75 - 1.25\psi$ and $1.25 + 1.25\psi$ and meet at $\psi = 0.2$, $p = 1.50$, the same price M&R use for the reversed-probability case (p. 143). The experimental magnitudes are Varian's (pp. 582–583).

Answer. Fairness and vengeance act as built-in commitment devices. A conscience makes promises credible and protects specific investments (guilt above 1 removes grabbing in Table 5.1); a taste for revenge makes threats credible and gives responders bargaining power, which explains offers of about 45 cents instead of the SPNE near zero. Honesty relaxes incentive-compatibility constraints (a lying cost of \$0.20 restores efficient trade in M&R's example). The cost is that privately known or self-serving notions of fairness bring new disagreement, and the preferences help only if they are observable and cannot be mimicked.

21 M&R Analytical Problem 5.4

Exam · block B3

Source: M&R Analytical Problem 5.4, page 165 (statement in the book; not reproduced here).

What is asked. A used-car seller and a buyer each hold private, independent values (seller \$250 or \$750, buyer \$500 or \$1,000, with the probabilities indexed by x). Say when trade is efficient and how likely that is; derive the two honest-reporting conditions on the mediator's price p for the reports (\$250, \$1,000); find the x at which both bind; and explain why a lower x makes trade more often efficient yet efficient trade unattainable.

Method. The mediator and incentive-compatibility method of M&R Ch. 5, "Bargaining over a Sale" (pp. 140–143, Tables 5.2–5.3): fix the prices that participation pins down, then write one honesty constraint per "tempted" type (high-value buyer, low-value seller), each computed assuming the other side reports honestly. By the revelation principle (M&R n. 8, pp. 142–143), if no honest-reporting mechanism yields efficient trade, no bargaining procedure does.

Solution. (a) When should trade occur? Trade is efficient whenever the buyer's value exceeds the seller's. The seller is worth \$250 with probability $1-x$ and \$750 with probability x; the buyer is worth \$500 with probability x and \$1,000 with probability $1-x$.

	Seller \$250 (prob. $1-x$)	Seller \$750 (prob. x)
Buyer \$500 (prob. x)	Trade (gain 250)	No trade
Buyer \$1,000 (prob. $1-x$)	Trade (gain 750)	Trade (gain 250)

Only the pair (buyer \$500, seller \$750) should not trade, so

$$\Pr(\text{trade is efficient}) = 1 - x \cdot x = 1 - x^2,$$

which *falls* as x rises. The full-information expected gain is $x(1-x)250+(1-x)^2750+(1-x)x\,250 = (1-x)(750-250x)$; for example \$560 at $x = 0.2$.

(b) The mediator's price table. As in M&R's Table 5.3, a seller who claims \$750 must receive at least \$750 when she trades (a true \$750 seller would otherwise refuse), and a buyer who claims \$500 cannot be charged more than \$500. Both honesty constraints below improve when the (seller \$250, buyer \$500) price is higher and the (seller \$750, buyer \$1,000) price is lower, so the mediator's best choice is to set them at these participation limits:

	Buyer reports \$500	Buyer reports \$1,000
Seller reports \$250	price \$500	price p
Seller reports \$750	no trade	price \$750

This choice loses no generality: if no p works with \$500 and \$750, no p works with any other admissible prices either.

(c) Seller worth \$250 (gains measured relative to keeping a car worth \$250 to her; the buyer reports honestly). Honest: she sells at \$500 with probability x and at p with probability $1-x$. Claiming \$750: no sale against a \$500 buyer (gain 0), a sale at \$750 against a \$1,000 buyer.

$$\underbrace{x(500-250)+(1-x)(p-250)}_{\text{honest}} \;\geq\; \underbrace{(1-x)(750-250)}_{\text{claim \$750}} \quad\Longleftrightarrow\quad p \;\geq\; 750 - \frac{250\,x}{1-x}. \qquad (\text{IC}_S)$$

(d) Buyer worth \$1,000. Honest: he pays \$750 with probability x and p with probability $1-x$. Claiming \$500: no trade with a \$750 seller, a purchase at \$500 from a \$250 seller.

$$\underbrace{x(1000-750)+(1-x)(1000-p)}_{\text{honest}} \;\geq\; \underbrace{(1-x)(1000-500)}_{\text{claim \$500}} \quad\Longleftrightarrow\quad p \;\leq\; 500+\frac{250\,x}{1-x}. \qquad (\mathrm{IC}_B)$$

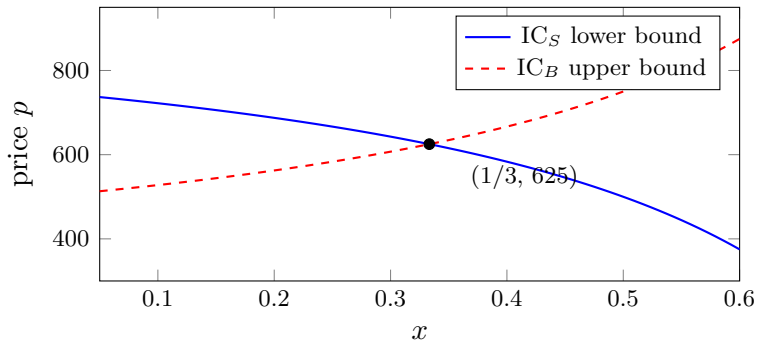
The other two types are not tempted once these hold. A \$500 buyer who claims \$1,000 gains $-250x+(1-x)(500-p)\leq 0$ iff $p\geq 500-250x/(1-x)$, implied by IC_S; a \$750 seller who claims \$250 gains $-250x+(1-x)(p-750)\leq 0$ iff $p\leq 750+250x/(1-x)$, implied by IC_B. Participation in the middle cell only needs $250\leq p\leq 1000$.

(e) Both constraints as equalities. With $k\equiv 250x/(1-x)$, a price exists iff $750-k\leq 500+k$, i.e. $k\geq 125$, i.e. $x/(1-x)\geq 1/2$:

$$\boxed{x^*=\tfrac{1}{3}}, \qquad k=125, \qquad p=625.$$

At $x=1/3$ the only honest price is \$625, the midpoint of \$250 and \$1,000. For $x>1/3$ there is an interval of prices; for $x<1/3$ there is none.

x	Pr(trade efficient) $=1-x^2$	k	IC_S: $p\geq$	IC_B: $p\leq$	efficient trade attainable?
0.10	0.99	27.78	722.22	527.78	no
0.20	0.96	62.50	687.50	562.50	no
1/3	0.89	125.00	625.00	625.00	yes, only $p=625$
0.50	0.75	250.00	500.00	750.00	yes, $p\in[500,750]$
0.80	0.36	1000.00	−250	1500	yes, $p\in[250,1000]$



Honest prices exist to the right of $x=1/3$, where the lower bound lies below the upper bound.

(f) Why a lower x helps efficiency but defeats it. A lower x makes the no-gain pair rarer, so trade is efficient more often ($1-x^2$ rises). But the same x is the probability that a lie *costs* the liar a trade: a \$1,000 buyer who claims \$500 loses the sale only if the seller is worth \$750 (probability x), and a \$250 seller who claims \$750 loses it only if the buyer is worth \$500 (probability x). The *gain* from lying is a better price in the likely cell, which occurs with probability $1-x$. As x falls, lying becomes almost free, so the rent each tempted type must be promised for honesty approaches the full \$500 it could grab by lying. Both rents must be paid out of the \$750 surplus of the (\$250, \$1,000) cell: $(p-250)+(1000-p)=750\geq 2(500-k)$, which is exactly $x\geq 1/3$. Below that there is "simply not enough surplus to go around" (M&R, p. 141). A mediator who maximizes expected gains must then sacrifice some efficient trades, e.g. by trading with probability below one in the cells the liars rely on, which is the incentive-efficiency logic of M&R pp. 143–145.

Interpretation. This is M&R's Table 5.2–5.3 example in disguise: rescaling values by (value − 250)/250 turns \$250/\$750 and \$500/\$1,000 into \$0/\$2 and \$1/\$3. Private information is a source of inefficiency that no amount of goodwill removes: the feasible set shrinks once honest reporting must be paid for with informational rents (M&R n. 7, p. 141). Efficient bargaining is easiest precisely when each side's lie is likely to kill a deal, that is, when the opponent is probably a type with whom a misreport destroys the trade.

Verification. Under the rescaling above, $x = 0.2$ reproduces M&R's bounds: $250 + 250 \times 1.75 = 687.5$ and $250 + 250 \times 1.25 = 562.5$, as in the table; $x = 0.8$ is M&R's reversed case, and their price $p = 1.50$ maps to \$625, inside $[250, 1000]$. `check_D.py` solves IC_S and IC_B symbolically, finds $x^* = 1/3$ and $p = 625$, recomputes every row of the table and checks that the low types' constraints are implied.

Answer. Trade is efficient except when the buyer is worth \$500 and the seller \$750, so with probability $1 - x^2$. With prices \$500 and \$750 in the other cells, honesty requires $p \geq 750 - 250x/(1-x)$ (seller) and $p \leq 500 + 250x/(1-x)$ (buyer). Both bind at $x = 1/3$, $p = \$625$. For $x < 1/3$ trade is more often efficient, but lies are almost costless, the rents needed for honesty exceed the \$750 surplus, and efficient trade is unattainable.

22 M&R Analytical Problem 5.5

Exam · block B3

Source: M&R Analytical Problem 5.5, page 165 (statement in the book; not reproduced here).

What is asked. Only the seller is informed: she values the good at \$0 (probability x) or \$10, and the buyer's value is the seller's plus a known premium. Show that with a \$10 premium a single price of \$10 supports honest reports and trade in every state; then, with a \$5 premium, decide whether report-contingent prices can make the seller honest while both sides are willing to trade.

Method. M&R's mediator and incentive-compatibility method (Ch. 5, pp. 141–143), here with one-sided private information: a price schedule $p(\hat{s})$ depending on the seller's report $\hat{s} \in \{0, 10\}$, honesty constraints for the two seller types, and participation constraints for both seller types and for the buyer, who learns the state from an honest report. Trade probabilities below one are the screening device of M&R's incentive-efficiency discussion (pp. 143–145).

Solution. Write $s \in \{0, 10\}$ for the seller's value and $s + g$ for the buyer's, with $g = 10$ or $g = 5$. Trade is efficient in both states because $g > 0$.

(a) Premium \$10: one price of \$10 always. Let $p(0) = p(10) = 10$ and trade always.

- *Honesty.* The report changes neither the price nor whether trade occurs, so the seller cannot gain by misreporting; honest reporting is (weakly) optimal for both types.
- *Seller participation.* Type 0 gains $10 - 0 = 10 \geq 0$; type 10 gains $10 - 10 = 0$, so she is willing to sell.
- *Buyer participation.* When $s = 0$ the good is worth \$10 to him and he pays \$10 (gain 0); when $s = 10$ it is worth \$20 (gain 10). He is willing to buy in each state, even if he could infer the state, so the scheme survives ex post.

State	Price	Seller's gain	Buyer's gain
$s = 0$ (prob. x)	10	10	0
$s = 10$ (prob. $1 - x$)	10	0	10

Efficient trade occurs in every state, with honest revelation. (Any constant price in $[10,\ 20 - 10x]$ also works if the buyer only needs a non-negative *expected* gain, since $x(10-p)+(1-x)(20-p) = 20 - 10x - p$; \$10 is the price that also works state by state.)

(b) Premium \$5: is there an honest schedule with trade in both states? Now the buyer is worth \$5 when $s = 0$ and \$15 when $s = 10$. Suppose trade always occurs at $p(0)$ and $p(10)$.

1. *Honesty of type 0*: $p(0) \geq p(10)$ (she sells either way, so only the price matters).
2. *Honesty of type 10*: $p(10) - 10 \geq p(0) - 10$, i.e. $p(10) \geq p(0)$. Together: $p(0) = p(10) \equiv P$. When the report cannot change whether trade happens, it cannot change the price either.
3. *Participation of type 10*: $P \geq 10$.
4. *Buyer's willingness to pay*: after an honest report of 0 the buyer knows the good is worth \$5 to him, so he buys only if $P \leq 5$.

Conditions 3 and 4 are incompatible: **no such prices exist**. More generally, trade in all states with honesty forces one pooled price that must lie in $[10,\ 0 + g]$, which is non-empty iff $g \geq 10$: the premium must be at least the gap between the seller's two values. That is exactly why the \$10 case works and the \$5 case fails.

What can be achieved instead. Honesty is restored by making the claim "\$10" costly through a lower chance of trade. Let trade occur for sure at $p(0) \leq 5$ after report 0, and with probability q at $p(10) \geq 10$ after report 10. Type 0 stays honest iff $p(0) \geq q\,p(10)$; with $p(0) = 5$ and $p(10) = 10$ this allows any $q \leq 1/2$. Type 10 does not mimic type 0, since selling at \$5 loses \$5; the buyer gains $5 - 5 = 0$ and $15 - 10 = 5$ in the two states. The best such scheme trades with probability $q = 1/2$ when the good is worth \$10 to the seller, and the expected loss against full efficiency is $(1 - x) \cdot \frac{1}{2} \cdot 5 = 2.5\,(1 - x)$.

Caveat on timing. If the buyer had to commit to a single price before anything is revealed and judged it by his expected gain, one price $P \in [10,\ 15 - 10x]$ would do, which is possible iff $x \leq 1/2$. Once prices depend on the report, though, the report reveals the state, and the buyer's willingness to trade must hold state by state, as above.

Interpretation. Private information does not need two-sided uncertainty to destroy efficient trade. With *interdependent* values the buyer's value moves with the seller's information. The low-cost seller always wants to be paid like a high-cost one, so honesty pools the prices. The buyer will not pay the high-cost price for a good he then knows is worth little. When the gains from trade are large ($g \geq 10$) a single price covers everybody; when they are small, the informed party's informational rent cannot be paid without giving up trade, and the incentive-efficient arrangement rations trade with the high-value seller. This is the logic of Akerlof's lemons model, which M&R discuss under adverse selection in the same chapter, in bargaining form.

Verification. Each constraint is checked by substitution in `check_D.py`: seller and buyer gains in the \$10 case; the condition $10 \leq P \leq g$, which holds iff $g \geq 10$; and $q = 1/2$ from $5 = q \times 10$, the buyer's gains (0 and 5) and type 10's loss ($5 - 10 < 0$) in the \$5 case; the interim range closes at $x = 1/2$.

Answer. With a \$10 premium, a constant price of \$10 and trade always is honest (reports change nothing) and individually rational for both seller types and for the buyer in each state. With a \$5 premium, no: honesty with trade in both states forces $p(0) = p(10) = P$, the \$10 seller needs $P \geq 10$, and a buyer who learns the good is worth \$5 pays at most \$5. Efficient trade with honesty needs a premium of at least \$10. The best honest scheme trades at \$5 after report 0 and with probability 1/2 at \$10 after report 10.

23 M&R Food for Thought 2.5

Exam · block B3

Source: M&R Food for Thought 2.5, page 52 (statement in the book; not reproduced here).

What is asked. Assess how specific a cable-television company's investment in wiring a community is, and predict the arrangements the company would seek with local communities over the pricing and taxation of its service.

Method. Transaction attributes, asset specificity first (M&R Ch. 2, pp. 30–31, and summary, p. 49); imperfect commitment and opportunistic renegotiation (p. 30); specific and cospecialized assets, the hold-up problem and the contractual remedies observed in coal supply (M&R Ch. 5, "Investments and Specific Assets", pp. 134–139); quasi-rents (M&R Ch. 8, p. 269); investments in bargaining position (M&R p. 149).

Solution. **(a) How specific is the investment? Very.** M&R measure specificity, when value maximization applies, as "the percentage of investment value that is lost when the asset is used outside the specific setting or relationship" (p. 135). For a cable network:

- *Site specificity.* Cable strung on poles or buried along a town's streets to its households cannot be moved to another town. The bulk of the cost is installation (trenching, pole attachments, drops to each house, engineering), which has no salvage value; only the head-end electronics and some equipment are redeployable.
- *Dedicated to one counterparty.* The network serves only that community, and the firm's access to streets, poles and subscribers runs through a franchise that the local government grants, renews and can regulate and tax.
- *Durable.* The cable lasts many years, so the investment is recovered only over a long horizon, during which conditions (technology, costs, local politics) change.

So most of the investment would be lost if the relationship with the community ended: the specificity is close to 100% for the installed plant. The specificity is *two-sided*, though not symmetric: once one network is in place, duplicating it is wasteful (cable distribution is a natural monopoly), so the community also has few alternatives and the firm could exploit its local monopoly. Ex ante, several firms can compete for the franchise; ex post, firm and community are locked in a bilateral relationship.

(b) The hazard: hold-up. After the cable is sunk, the community can change the terms: cap subscriber prices, raise franchise fees or taxes, impose new service obligations, or threaten not to renew. The firm keeps operating as long as revenue covers its *avoidable* costs, so the community can appropriate the firm's quasi-rent, the excess of revenue over what the firm needs to stay rather than exit (M&R p. 269). A firm that foresees this underinvests or does not enter at all. A minimal example: the network costs $K = 100$ (fully specific) and generates operating returns worth $R = 150$ in present value. Building is efficient ($150 > 100$). If, after the investment, the

returns are split 50/50 by renegotiation with the community, the firm gets $75 < 100$ and does not build: 50 of value is lost although nobody behaves irrationally. The danger runs both ways: a firm with a captive market can also hold up subscribers with high prices or poor service. This is M&R's mine-mouth plant problem (pp. 135–137), with the community in the role of the utility.

A community is also a government: it can change taxes and regulations unilaterally, and its promises are hard to enforce against it. Imperfect commitment (M&R p. 30) is therefore particularly severe here.

(c) Arrangements to expect. Because the investment is large and specific, the parties "seek to organize in ways that safeguard those investments" (M&R p. 49). Before investing, the company would want a long-term, detailed franchise contract with the community, of the kind Joskow found for coal (M&R p. 137):

(i) *Long duration*, enough to recover the investment, with renewal procedures that presume renewal for satisfactory performance, or compensation for the network (e.g. purchase at an agreed valuation) if the franchise is not renewed or is given to a rival.

(ii) *Pricing rules fixed in advance*: an initial rate schedule with escalator clauses tied to cost or price indices, or a price-cap formula, rather than rates set at the community's discretion after the cable is laid.

(iii) *Taxation fixed by formula*: a franchise fee set as a stated percentage of gross revenue, with a ceiling, instead of discretionary taxes that could be raised later.

(iv) *Exclusivity or protection against overbuilding* for a period, so that the community cannot extract terms by threatening to license a competitor once the firm has sunk its cost.

(v) *Obligations on the firm*, protecting the community against the monopoly side of the lock-in: build-out to all neighbourhoods, service-quality standards, regulated rates for a basic tier, public-access channels.

(vi) *Dispute resolution* by arbitration or an outside regulator, and where possible commitments at a higher level of government that limit what local authorities can change later.

The alternative remedy is the one M&R report for GM and Fisher Body (p. 137): unified ownership. Here that means municipal ownership of the network, with the operation contracted out, which removes the firm's exposure but brings its own incentive problems.

(d) Investments in bargaining position. Since the terms are renegotiated at each renewal, both sides have an incentive to spend resources that shift the division without creating value (M&R p. 149): the community may cultivate a credible threat to license a second operator or build a municipal system; the firm may lobby, litigate, or design its network to raise the community's switching costs. Such spending is pure waste in total-value terms, and good contracts reduce it by fixing the renewal terms in advance.

Interpretation. This item connects the Ch. 2 vocabulary (asset specificity, imperfect commitment) to the hold-up logic developed later in the course. Under the Coase benchmark the efficient network would be built whatever the bargaining power; with specific assets and incomplete, imperfectly enforceable contracts, the threat of ex-post renegotiation distorts the ex-ante investment. The transaction cost approach predicts the governance structure that maximizes total value net of those costs: long-term contracts with pricing formulas, protected tax treatment and renewal guarantees, or integration. (For a classic study of cable franchise bidding along these lines, see Williamson, "Franchise Bidding for Natural Monopolies—in General and with Respect to CATV", *Bell Journal of Economics*, 1976.)

Verification. Internal consistency: the efficient decision compares $R = 150$ with $K = 100$; the firm compares its post-renegotiation share $R/2 = 75$ with K. Any ex-ante agreement guaranteeing the firm at least 100 out of 150 (e.g. a fee formula capping the community's take at 50) restores the investment, consistent with the remedies in (c) (`check_C.py`).

Answer. The investment is highly specific (site-specific, durable, dedicated to one community; most of its value is lost outside the relationship), which exposes the company to hold-up through price controls, taxes or non-renewal once the cable is sunk. Expect long-term franchise agreements signed before investment: long duration and renewal or buy-out guarantees, rate formulas with escalators or price caps, franchise fees fixed as a capped share of revenue, exclusivity, and service obligations protecting the community; or, as the alternative, municipal ownership.

Tier 3 — Advanced

24 M&R Mathematical Exercise 4.2

Advanced · block A3

Source: M&R Mathematical Exercise 4.2, page 120 (statement in the book; not reproduced here).

What is asked. A product can be launched only when three departments are all ready; earlier readiness raises post-launch profit but each department pays a linear cost for speed. Find the optimal launch date, the profit when all departments are mistimed together by one month, the profit when only department 2 is mistimed by one month, and say whether the launch date is a design decision.

Method. Chapter 4's model of design decisions: "The Variety of Coordination Problems and Solutions" (pp. 90–93, design attributes p. 91), "Planning with Design Attributes" (pp. 103–106, Figure 4.2 p. 105) and the Mathematical Appendix (pp. 121–124): revenue depends on the date of the *last* department, $R[\max_n t^n]$ (Eq. 4.3, p. 121); at an optimum all dates are equal and marginal benefit of a speed-up equals the sum of marginal costs (Eqs. 4.5–4.6, p. 122); a common error costs nothing to first order, a misfit costs a first-order amount (Eq. 4.7, pp. 123–124).

Solution. Write t_n for the date department n is told to be ready, $c_1 = 3$, $c_2 = 4$, $c_3 = 5$ for the cost per month of speeding it up, and $C = c_1 + c_2 + c_3 = 12$. Production starts at $T = \max\{t_1, t_2, t_3\}$, so for dates below 12

$$\pi(t_1, t_2, t_3) \;=\; 144 - T^2 - \sum_{n=1}^{3} c_n\,(12 - t_n), \qquad T = \max_n t_n.$$

For $t \geq 12$ the launch earns nothing, and no department is paid to get ready, so $\pi = 0$.

(1) Optimal date. No department should be ready before the last one: lowering t_n below T raises cost by c_n per month and leaves revenue unchanged. Hence at an optimum $t_1 = t_2 = t_3 = t$ (the equality constraints of Eq. 4.5, p. 122) and

$$\pi(t) = 144 - t^2 - 12(12 - t) = 12t - t^2, \qquad \pi'(t) = 12 - 2t = 0 \implies t^* = 6,$$

with $\pi'' = -2 < 0$, so the stationary point is the global maximum on $[0, 12)$; the corners give $\pi(0) = 0$ and $\pi = 0$ for $t \geq 12$. At $t^* = 6$: revenue $144 - 36 = 108$, readiness cost $12 \times 6 = 72$,

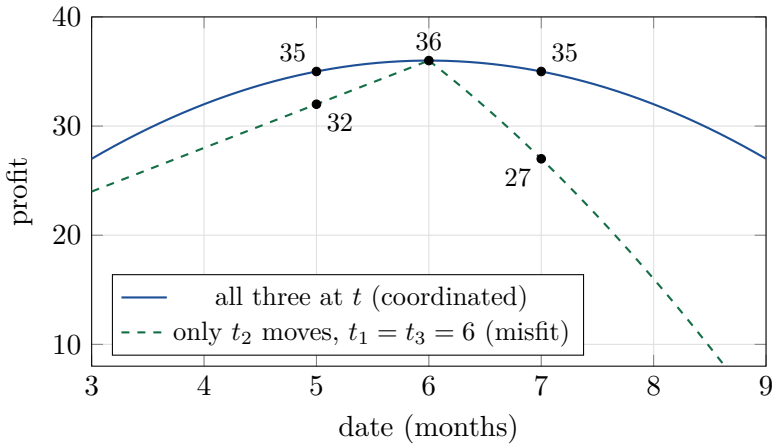
profit $\pi^* = 36$. The first-order condition reads $2t^* = c_1 + c_2 + c_3$: the marginal benefit of launching a month sooner equals the *sum* of the departments' marginal costs of speeding up — the third adding-up condition in Eq. 4.6 and the crossing in Figure 4.2 (p. 105).

(2) All three mistimed together. $\pi(5) = 60 - 25 = 35$ and $\pi(7) = 84 - 49 = 35$. Each costs one unit. In general $\pi(6 + \varepsilon) = 36 - \varepsilon^2$: a coordinated error is a second-order loss, because at t^* the marginal benefit and the summed marginal cost cancel (p. 123).

(3) Department 2 alone mistimed, departments 1 and 3 at $t^* = 6$.

- *Department 2 one month early* ($t_2 = 5$). Launch still at $T = 6$: revenue 108, cost $3 \cdot 6 + 4 \cdot 7 + 5 \cdot 6 = 76$, profit 32. Loss $4 = c_2 \times 1$: department 2 pays for speed nobody can use.
- *Department 2 one month late* ($t_2 = 7$). Launch slips to $T = 7$: revenue $144 - 49 = 95$, cost $3 \cdot 6 + 4 \cdot 5 + 5 \cdot 6 = 68$, profit 27. Loss 9: departments 1 and 3 paid for a month of speed that is wasted ($3 + 5 = 8$), plus the second-order term $\varepsilon^2 = 1$. Equivalently, $27 = \pi(7) - (c_1 + c_3) = 35 - 8$.

In general, with $t_m = t^*$ for $m \neq n$, a misfit of size ε costs $c_n\varepsilon$ if department n is early and $(C - c_n)\varepsilon + \varepsilon^2$ if it is late. The profit is kinked at t^*: as a function of t_2 its left slope is $c_2 = 4$ and its right slope is $-(c_1 + c_3) = -8$ — the left- and right-hand derivatives of Eq. 4.7 (p. 123), both non-zero.



(4) A design decision? Yes; both attributes of p. 91 hold. (i) *A priori information about the form of the optimum*: before knowing any cost or revenue numbers, the coordinator knows that every department must be ready on the same date, because revenue depends only on the last date (Eq. 4.3, p. 121; Eq. 4.5, p. 122). (ii) *Misfits cost more than a wrong common design*: a one-month error costs 1 when all move together but 4 or 9 when one department is out of step — second-order against first-order losses (pp. 123–124).

The organizational corollary (pp. 104–106): the centre should announce the *date* $t^* = 6$, one number, and check it with the marginal benefit $2t$ and the three marginal costs (four numbers, cf. the "three numbers" for two suppliers on p. 105). Prices would do badly. If the centre paid department n a price p per month of earliness, its payoff $(p - c_n)(12 - t_n)$ is linear in t_n: it would choose $t_n = 0$ if $p > c_n$, $t_n = 12$ if $p < c_n$, and be indifferent if $p = c_n$. No single price synchronizes the three; department-specific prices equal to c_n leave each indifferent, so the date must still be dictated. This is the brittleness and communication argument of p. 104 in its sharpest form.

Interpretation. Getting the parts to fit matters more than getting the common target exactly right. This is the coxswain's call (p. 91) and the new-car launch date (p. 104) in numbers: centralize the design variable (the date), and leave each department free to choose *how* to be ready (p. 115). It is also why a planner who is unsure of t^* should still insist that everyone works to the same date: a wrong common date costs little, a broken synchronization costs a lot.

Verification. (a) The first-order condition $2t^* = C$ gives $t^* = C/2 = 6$. Check: $\pi(5.9) = 35.99$ and $\pi(6.1) = 35.99$, both below 36. (b) The misfit losses add up: delaying t_2 by one month changes revenue by $-(49 - 36) = -13$ and saves department 2 4, net -9. Bringing t_2 forward costs 4 and changes no revenue. (c) Limit case: if department 2 were the only department ($c_1 = c_3 = 0$), being late would cost only ε^2; the first-order misfit loss $(C - c_n)\varepsilon$ appears only because the others have already paid to be ready. (d) `check_A.py` verifies t^*, every profit $(36, 35, 35, 32, 27)$, the kink slopes 4 and -8, the general loss formulas for each department, and the corner response to a price.

Answer. (1) $t^* = 6$ months, profit 36 (revenue 108 minus readiness cost 72), where $2t^* = 3{+}4{+}5$. (2) All one month early or late: profit 35 each (loss 1, second order). (3) Department 2 early: 32 (loss 4); department 2 late: 27 (loss 9); misfits are first-order losses. (4) Yes: the optimum is known to require equal dates, and misfits cost far more than a common timing error, so the centre should announce the date rather than prices.

25 Cabral 3.20 Productivity (applied)

Advanced · block A2

Source: Cabral 3.20 Productivity (applied), page 64 (statement in the book; not reproduced here).

What is asked. Obtain firm-level data on outputs and inputs (quantities and prices) and estimate each firm's total factor productivity with the method of §3.1 (Equation 3.2).

Method. TFP as the multiplicative coefficient of a Cobb-Douglas technology common to the industry, recovered as a residual, with the exponents measured by cost shares (Cabral §3.1, "Productivity" and "Estimating total factor productivity", pp. 39–40, eqs. 3.1–3.2; Exercise 3.15 for the share result). The benchmark for dispersion is the evidence in §4.2 (p. 75): within US four-digit industries the 90th-percentile plant has close to twice the TFP of the 10th, and close to five times in China and India. Course link: "organization matters" predicts large, persistent productivity gaps between firms with the same inputs (§3.5, pp. 56–59; M&R Ch. 1).

Solution.

(a) The method. Assume, as Cabral does (p. 39), that (i) all firms face the same output price, (ii) each firm i has $q_i = \omega_i K_i^{\alpha} L_i^{\beta}$, and (iii) α and β are common while ω_i differs. Taking logs,

$$\ln \omega_i = \ln q_i - \alpha \ln K_i - \beta \ln L_i. \tag{3.2}$$

A price-taking, profit-maximizing firm sets $\alpha\, pq_i = rK_i$ and $\beta\, pq_i = wL_i$ (the first-order conditions of $\max\ p\,\omega_i K^{\alpha} L^{\beta} - rK - wL$), so the exponents equal the revenue shares of capital and labour costs:

$$\hat{\alpha} = \overline{\left(\frac{rK}{pq}\right)}, \qquad \hat{\beta} = \overline{\left(\frac{wL}{pq}\right)},$$

averaged across firms. Plug $\hat{\alpha}$, $\hat{\beta}$ into (3.2) to get each $\ln \hat{\omega}_i$, normalize by the industry mean, and summarize the dispersion by the standard deviation of $\ln \hat{\omega}$ and the 90/10 ratio $\exp(p_{90} - p_{10})$.

(b) Measurement issues.

(i) *Labour productivity is the wrong measure.* Revenue per worker rises with capital even when two firms share the same technology (p. 39). Under profit maximization it can say almost nothing: $pq/L = w/\beta$, so it reflects the wage, not efficiency.

(ii) *Quantity versus revenue.* Most datasets report revenue. Dividing by an industry price index measures quantity only if all firms charge the same price (assumption (i)). With firm-specific prices, "revenue TFP" mixes efficiency with market power and demand, the productivity-versus-profitability distinction of Foster, Haltiwanger and Syverson (2008), which Cabral cites in Chapter 4.

(iii) *Shares and market power.* With a markup, revenue shares understate the exponents; using shares of total cost (under constant returns) is the usual correction.

(iv) *Capital and its user cost* are hard to measure (book values, depreciation, the rate r).

(v) *Regression instead of shares.* Estimating α, β by OLS of $\ln q$ on $\ln K$, $\ln L$ is biased, because firms choose inputs knowing their own ω_i: more productive firms use more inputs, so OLS credits inputs with part of the productivity. Olley and Pakes (1996), also cited by Cabral, address this.

(c) A worked run on SYNTHETIC data. No real dataset is shipped with these notes, so the script `tfp_example.py` (in this folder) demonstrates the method on **simulated firms**. It draws 500 price-taking, profit-maximizing firms with true $\alpha = 0.30$, $\beta = 0.50$ and log-TFP standard deviation 0.27 (chosen so that the true 90/10 ratio is about 2, the US benchmark of p. 75), firm-specific input prices, small input-choice errors and output measurement error (seed 2017). **The numbers below describe these simulated firms and are not evidence about any real industry.** Output of `python tfp_example.py`:

```
SYNTHETIC DATA (seed 2017), 500 simulated firms; true alpha=0.30, beta=0.50
cost-share estimates: alpha=0.300  beta=0.502
log TFP: sd=0.274   90/10 TFP ratio=2.02
naive OLS of ln q on ln K, ln L: alpha=0.491  beta=0.495
corr(log labour productivity, log TFP) = 0.057
true log TFP: sd=0.276   90/10 ratio=2.01   corr(estimated, true)=0.983
```

Reading: the cost-share method recovers the exponents and the TFP ranking (correlation 0.98 with the true values) and the dispersion (90/10 ratio 2.02 against 2.01). The naive OLS regression overstates α (0.49 against 0.30) because input choices respond to ω_i, illustrating issue (v). Labour productivity is nearly uncorrelated with TFP (0.06), illustrating issue (i).

(d) Using real data. Save a CSV with one row per firm (same industry and year) and columns `q, K, L, p, r, w`: output (quantity, or revenue deflated by an industry price index), capital, labour, output price, user cost of capital and wage. Then run `python tfp_example.py firms.csv`; the same estimates are printed, labelled as CSV data. Typical sources are manufacturing censuses and annual industrial surveys with establishment microdata (for Brazil, IBGE's PIA through a secure data room) or commercial balance-sheet panels; with revenue data, issue (ii) applies and the result is revenue TFP.

Interpretation. On real data the method typically finds what §4.2 reports: large dispersion (about 2:1 between the 90th and 10th percentile within narrow US industries) that persists over time. Firms with the same inputs produce very different outputs. That residual is where the course puts organization: management practices, incentive systems and the fit among them (§3.5, p. 58; M&R Ch. 1). TFP measures the size of the gap; it does not by itself show that organization causes it, which is why Cabral turns to field experiments for causality (p. 58).

Verification. (1) With the true α, β, the script's estimates are within 0.002 of the truth, and the script asserts $|\hat{\alpha} - 0.3| < 0.02$, $|\hat{\beta} - 0.5| < 0.02$ and a correlation above 0.9. (2) Limit case: with no TFP dispersion the residual would contain only measurement and input-choice error (s.d. about 0.06 in the simulation, which alone gives a 90/10 ratio of about $e^{2\times 1.2816\times 0.058} \approx 1.16$), so a measured 90/10 ratio near 2 cannot come from noise of that size. (3) Analytic check of issue (i): under the first-order conditions $pq/L = w/\beta$ exactly, so the small correlation in the output is the expected one, not a coding error. (4) Running the script on a CSV written from the synthetic firms reproduces the same estimates, confirming the CSV path.

Answer. Estimate α, β as average revenue shares of capital and labour costs, then $\ln \hat{\omega}_i = \ln q_i - \hat{\alpha} \ln K_i - \hat{\beta} \ln L_i$ (eq. 3.2), and report the spread (s.d., 90/10 ratio). Watch for revenue versus quantity, markups, capital measurement and the simultaneity bias of OLS. On *synthetic* data the script recovers $\hat{\alpha} = 0.300$, $\hat{\beta} = 0.502$ and a 90/10 ratio of 2.02 (true 2.01). Real data typically show ratios near 2 (US) to 5 (China, India), the dispersion that "organization matters" predicts.

26 M&R Mathematical Exercise 2.3

Advanced · block B1

Source: M&R Mathematical Exercise 2.3, page 53 (statement in the book; not reproduced here).

What is asked. Prove the value maximization principle for N people with quasilinear utilities who share the net profit of a decision: an allocation is efficient exactly when its decision maximizes total value.

Method. Definitions of feasibility and Pareto efficiency; the identity "sum of utilities = total value" under quasilinearity; a direct argument for sufficiency and the equal-split transfer construction of the book's hint for necessity (M&R pp. 35–37, Figure 2.1; notes, Part II, "The value maximization principle"). The remark on wealth effects follows M&R n. 10, p. 37.

Solution. Setting. Individuals $n = 1, \ldots, N$ have utilities $u_n(x_n, y) = x_n + v_n(y)$, where y ranges over an arbitrary set Y of decisions and $x_n \in \mathbb{R}$ is n's cash receipt (a payment if negative). Decision y generates net profit $P(y)$. An allocation $(y, x_1, \ldots, x_N)$ is *feasible* if $\sum_n x_n = P(y)$. A feasible allocation A' *Pareto dominates* A if $u_n(A') \geq u_n(A)$ for all n with strict inequality for some n; A is *efficient* if no feasible allocation dominates it. Define total value

$$W(y) \;\equiv\; P(y) + \sum_{n=1}^{N} v_n(y).$$

Claim. *A feasible allocation $A = (y, x_1, \ldots, x_N)$ is efficient if and only if $W(y) \geq W(y')$ for every $y' \in Y$.*

Lemma (sum of utilities). For every feasible allocation,

$$\sum_n u_n(x_n, y) = \sum_n x_n + \sum_n v_n(y) = P(y) + \sum_n v_n(y) = W(y).$$

The sum of utilities depends on y only; transfers move utility between people one for one and never change the total (the 45° lines of M&R's Figure 2.1).

Proof of "if" (value maximizing ⇒ efficient). Let y maximize W and suppose, for a contradiction, that a feasible $A' = (y', x'_1, \ldots, x'_N)$ dominates A. Summing the N inequalities $u_n(A') \geq u_n(A)$,

at least one of them strict, and using the lemma for both allocations,

$$W(y') = \sum_n u_n(A') \;>\; \sum_n u_n(A) = W(y),$$

contradicting the maximality of $W(y)$. Hence no feasible allocation dominates A. □

Proof of "only if" (efficient ⇒ value maximizing). By contraposition: suppose y does *not* maximize W. Then there is $y^* \in Y$ with

$$G \;\equiv\; W(y^*) - W(y) > 0.$$

Following the hint, divide the gain equally. Define

$$x_n^* \;=\; x_n + v_n(y) - v_n(y^*) + \frac{G}{N}, \qquad n = 1, \ldots, N.$$

Everyone strictly gains: $u_n(x_n^*, y^*) = x_n^* + v_n(y^*) = x_n + v_n(y) + G/N > u_n(x_n, y)$ for every n.

The allocation is feasible: using $\sum_n x_n = P(y)$ and the definition of G,

$$\sum_n x_n^* = P(y) + \sum_n v_n(y) - \sum_n v_n(y^*) + G = W(y) - \sum_n v_n(y^*) + W(y^*) - W(y) = P(y^*).$$

So $(y^*, x_1^*, \ldots, x_N^*)$ is feasible and strictly preferred by all N people: A is not efficient. □

The second part proves more than the claim: every inefficient allocation is *strictly* Pareto dominated (all N people gain) by an allocation using a better y^*, and if y^* is chosen in $\arg\max W$ the dominating allocation is itself efficient. That is the second sentence of M&R's statement of the principle (p. 36). Note also that efficiency restricts y only: every transfer vector adding up to $P(y)$ is efficient when y is value maximizing.

Where "no wealth effects" enters. Two places.

(i) *The form* $x_n + v_n(y)$ (conditions (i)–(ii) of the definition, M&R p. 35; item 27 derives it). It makes utility transferable one for one, so that the lemma holds and the utility possibility set for each y is a 45° hyperplane.

(ii) *Affordability of the transfers* (condition (iii)). The "only if" construction may require some x_n^* to be large and negative: person n's payment rises by $v_n(y^*) - v_n(y) - G/N$, so whoever gains most from the switch pays for it. The proof takes $x_n \in \mathbb{R}$ unrestricted. If each person can pay at most his or her wealth, $x_n \geq -w_n$, the "if" direction survives (it constructs nothing), but the "only if" direction can fail. Example: $N = 2$, $Y = \{a, b\}$, $P \equiv 0$, $v_1(a) = 0$, $v_1(b) = 10$, $v_2(a) = 5$, $v_2(b) = 0$, and $w_1 = 1$. Then $W(a) = 5 < W(b) = 10$. The allocation $(a, 0, 0)$ is not value maximizing, yet it is efficient under the wealth constraint: allocations with $y = a$ only redistribute, and any allocation with $y = b$ gives person 2 $x_2 = -x_1 \leq w_1 = 1 < 5 = u_2(a, 0)$. Person 1 cannot afford to compensate person 2.

Remark: wealth effects (M&R n. 10, p. 37). With general utilities $U_n(x_n, y)$ increasing in money, an allocation that maximizes $\sum_n U_n$ over feasible allocations is still efficient (the "if" argument uses only that summing Pareto inequalities raises the sum). But the converse fails: there can be efficient allocations that do not maximize the sum. The reason is that the utility possibility frontier for a given y is no longer a straight line with slope -1: moving a dollar from one person to another changes their utilities by different amounts that depend on their wealth. Efficient points then correspond to maximizing *weighted* sums $\sum_n \lambda_n U_n$ with weights that vary along the frontier, and different points of the frontier can use different decisions y (the WTP/WTA example of item 7). "Total value" is a meaningful single index of efficiency only because quasilinearity fixes all weights at one.

Interpretation. This is the formal backbone of M&R Ch. 2: under no wealth effects, efficiency is equivalent to choosing the y with the largest total value, and distribution is handled entirely by the transfers. Combined with the hypothesis of efficient bargaining it gives the Coase theorem (p. 38). The affordability counterexample is exactly the limit M&R stress on p. 39: with very limited capital, value-creating changes can be blocked.

Verification. Beach example (items 16–17): W is maximized at $y^* = 4$; from the equal-share allocation at $\bar{y} = 3.75$ the construction gives each family $G/N = (5/32)/4 = 5/128$ more, with contributions summing to $4 = -P(4)$. Counterexample: $W(a) = 5$, $W(b) = 10$, and the largest u_2 reachable at b with $x_1 \geq -1$ is $1 < 5$ (`check_C.py`).

Answer. With $u_n = x_n + v_n(y)$ and $\sum_n x_n = P(y)$, the sum of utilities of any feasible allocation equals $W(y) = P(y) + \sum_n v_n(y)$. If y maximizes W, a Pareto improvement would raise W: contradiction. If it does not, some y^* has $G = W(y^*) - W(y) > 0$, and transfers $x_n^* = x_n + v_n(y) - v_n(y^*) + G/N$ are feasible and make everyone strictly better off. The necessity part needs transfers to be affordable; with wealth effects, value maximization is sufficient but not necessary for efficiency.

27 M&R Mathematical Exercise 2.4

Advanced · block B1

Source: M&R Mathematical Exercise 2.4, page 53 (statement in the book; not reproduced here).

What is asked. Show that if every switch between decisions has a cash equivalent that does not depend on the decision maker's other cash, and more money is preferred to less, then preferences are represented by cash plus a value function of the decision.

Method. Axiomatic derivation of quasilinear utility from "no wealth effects" (M&R pp. 35–36 and n. 7; notes, Part II, "Wealth effects and the value index"), following the hint: construct v from the compensation function C with a fixed reference decision, show indifference between (x, y) and $(x + v(y), \bar{y})$, then compare on the reference decision using monotonicity.

Solution. Setting and assumptions. The decision maker has a complete and transitive preference $\succsim$ over pairs (x, y), with $x \in \mathbb{R}$ a cash receipt (negative: a payment) and $y \in Y$ a decision. Write $\sim$ for indifference and $\succ$ for strict preference. The book's hypotheses are:

(A1) *Compensation exists.* For all $y, y' \in Y$ there is $C(y, y') \in \mathbb{R}$ with $(C(y, y'), y) \sim (0, y')$.

(A2) *It does not depend on other cash* (no wealth effects). For all $z \in \mathbb{R}$: $(z + C(y, y'), y) \sim (z, y')$.

(A3) *More money is better.* For every y: $(x, y) \succ (x', y) \iff x > x'$.

(A1) is the case $z = 0$ of (A2); both are kept to mirror the book.

Construction. Fix any reference decision $\bar{y} \in Y$ and define

$$v(y) \;\equiv\; C(\bar{y}, y), \qquad y \in Y,$$

the cash that, added to $\bar{y}$, makes the decision maker as well off as y with no cash.

Step 1: $(x, y) \sim (x + v(y), \bar{y})$ for all x, y. Apply (A2) with the pair $(\bar{y}, y)$ and $z = x$:

$$(x + C(\bar{y}, y), \bar{y}) \sim (x, y), \quad \text{i.e.} \quad (x + v(y), \bar{y}) \sim (x, y).$$

In words: having y is the same as having $\bar{y}$ plus the cash equivalent $v(y)$, whatever cash x comes with it. This is where (A2) is essential: without it the compensation needed at cash level x would be some $C_x(\bar{y}, y)$ depending on x.

Step 2: comparison of two allocations. Take any (x, y) and (x', y'). By Step 1, $(x, y) \sim (x + v(y), \bar{y})$ and $(x', y') \sim (x' + v(y'), \bar{y})$. By transitivity of $\succsim$ (indifference can replace either side of a comparison),

$$(x, y) \succ (x', y') \iff (x + v(y), \bar{y}) \succ (x' + v(y'), \bar{y}) \iff x + v(y) > x' + v(y'),$$

the last step by (A3) applied to the common decision $\bar{y}$. The same argument with $\sim$ (using (A3) in both directions, which gives $(x, \bar{y}) \sim (x', \bar{y}) \iff x = x'$) shows $(x, y) \sim (x', y') \iff x + v(y) = x' + v(y')$. Therefore

$$u(x, y) \;=\; x + v(y)$$

represents $\succsim$: $(x, y) \succsim (x', y') \iff u(x, y) \geq u(x', y')$.

The hint's version. If U is any utility function representing $\succsim$, Step 1 gives $U(x, y) = U(x + v(y), \bar{y})$ and $U(x', y') = U(x' + v(y'), \bar{y})$. By (A3), $t \mapsto U(t, \bar{y})$ is strictly increasing, so $U(x, y) > U(x', y') \iff x + v(y) > x' + v(y')$: every representation is a strictly increasing transformation of $x + v(y)$.

Consistency and uniqueness.

- $v(\bar{y}) = C(\bar{y}, \bar{y}) = 0$: by (A1), $(C(\bar{y}, \bar{y}), \bar{y}) \sim (0, \bar{y})$, and (A3) forces $C(\bar{y}, \bar{y}) = 0$.
- The compensation function is recovered from v: $(C(y, y'), y) \sim (0, y')$ means $C(y, y') + v(y) = v(y')$, i.e. $C(y, y') = v(y') - v(y)$, the same at every wealth level.
- v is unique up to an additive constant: if $x + w(y)$ also represents $\succsim$, Step 1 gives $x + v(y) + w(\bar{y}) = x + w(y)$, so $w = v + w(\bar{y})$. Changing the reference decision only shifts v by a constant.

Why the hypothesis bites. Cobb–Douglas preferences $U = x\,y$ (with $x, y > 0$) satisfy (A3) and have a compensating payment at every cash level $z > 0$, but violate (A2): the compensation solving $(z + C)\,y = z\,y'$ is $C = z\,(y' - y)/y$, which grows with the cash z already held. A richer person needs more cash to accept a worse y, a wealth effect, and no representation of the form $x + v(y)$ exists.

Domain caveat (condition (iii)). The proof uses $x + v(y)$ for arbitrary x; if cash is bounded below ($x \geq -w$), Step 1 requires $x + v(y)$ to stay in the domain, which is M&R's condition that the decision maker can afford the payments (p. 35). The representation is then valid on the set where those payments are feasible.

Interpretation. This exercise is the converse of the "why" given in the notes: quasilinear utility implies no wealth effects, and no wealth effects (plus monotonicity in money) imply quasilinear utility. Hence $v(y)$ is a cash-equivalent *value* of the decision, comparable across people, and the value maximization principle (item 26) and the Coase theorem rest on exactly these hypotheses.

Verification. For a quasilinear example $u = x + 2\sqrt{y}$ with $\bar{y} = 1$: $C(\bar{y}, y)$ solves $C + 2 = 2\sqrt{y}$, so $v(y) = 2\sqrt{y} - 2$, a constant shift of $2\sqrt{y}$, and $C(y, y') = 2\sqrt{y'} - 2\sqrt{y}$ is independent of z. For $U = xy$, C depends on z (`check_C.py`).

Answer. Set $v(y) = C(\bar{y}, y)$. Independence of the compensation from other cash gives $(x, y) \sim (x + v(y), \bar{y})$ for every x; comparing on the common decision $\bar{y}$ with "more money is better" gives $(x, y) \succsim (x', y') \iff x + v(y) \geq x' + v(y')$. So $x + v(y)$ represents the preferences, with v unique up to a constant and $C(y, y') = v(y') - v(y)$.

28 MWG Exercise 9.B.13

Advanced · block B2

Source: MWG Exercise 9.B.13, page 302–303 (statement in the book; not reproduced here).

What is asked. Exercise 9.B.13 asks for the SPNE of the infinite-horizon version of Exercise 9.B.7. That exercise is the alternating-offer game of Example 9.AA.1 in which, instead of discounting, each offer costs its maker $c < v$ (incurred even if no agreement is reached), solved for the finite horizon T and as $T \to \infty$. We solve 9.B.7 first, because it is the benchmark the infinite game is compared with.

Method. Backward induction on the finite game, as in MWG Example 9.AA.1 (pp. 296–297), with a responder who is indifferent accepting (MWG n. 16). For the infinite game we use stationary strategies and the Shaked–Sutton bounding argument of Example 9.AA.2 (p. 298), and show that it no longer pins down a unique outcome. The course notes (Part II, "Alternating offers") give the discounted benchmark $v_1(T) = v\,[1 - (-\delta)^T]/(1+\delta) \to v/(1+\delta)$.

Solution. Payoffs are *net*: a player's share of v minus c times the number of offers he or she has made. The proposer must make an offer in each period (as in Figure 9.AA.1).

(a) Finite horizon (Exercise 9.B.7). Let a_k be the continuation payoff, net of the current offer's cost, of the player who proposes when k periods remain, and set $a_0 = 0$ (after period T there is nothing).

- With k periods left, the responder can reject and become the proposer with $k-1$ periods left, worth a_{k-1} (her past costs are sunk). So she accepts iff offered at least a_{k-1}, and the proposer offers exactly that:

$$a_k = v - a_{k-1} - c, \qquad a_0 = 0.$$

- Making an offer that will be rejected is worse: it yields $-c + a_{k-2}$, and $a_k - (a_{k-2} - c) = v - (a_{k-1} + a_{k-2}) = c > 0$ for $k \geq 2$ (for $k = 1$ it yields $-c < v - c$).

The recursion gives $a_1 = v - c$, $a_2 = 0$, $a_3 = v - c$, …, that is $a_k = v - c$ for odd k and $a_k = 0$ for even k. In period t the number of remaining periods is $k = T - t + 1$, so the unique SPNE is:

Periods with k remaining	Proposer offers the responder	Responder accepts iff offered
k odd (same parity as T)	0 (keeps v, net $v - c$)	at least 0
k even	$v - c$ (keeps c, net 0)	at least $v - c$

Agreement is immediate. Player 1's period-1 proposal and the net payoffs are

$$T \text{ odd: } (v, 0), \quad (u_1, u_2) = (v - c,\ 0); \qquad T \text{ even: } (c,\ v - c), \quad (u_1, u_2) = (0,\ v - c).$$

In one formula, $u_1(T) = (v-c)\,[1-(-1)^T]/2$, the discounted formula of Example 9.AA.1 evaluated at $\delta = 1$ with v replaced by $v - c$. *The player who makes the last offer gets the entire surplus net of one offer cost*; the other gets nothing. **As $T \to \infty$ there is no limit**: $u_1(T)$ alternates between $v - c$ and 0.

What changes relative to Example 9.AA.1. With discounting the last mover's advantage shrinks geometrically, $v_1(T) = v[1 - (-\delta)^T]/(1+\delta)$, and both parities converge to $v/(1+\delta)$. A fixed cost of making offers does not shrink with distance from the deadline: a responder's threat

to reject is worth exactly the continuation a_{k-1}, so the parity of T decides everything at every horizon.

(b) Infinite horizon (Exercise 9.B.13). Stationary equilibria. Consider the profile $\sigma(a)$, with $a \in [0, v-c]$ and $b \equiv v-c-a$: player 1, when proposing, offers player 2 the amount b, and player 2 accepts iff offered at least b; player 2, when proposing, offers player 1 the amount a, and player 1 accepts iff offered at least a.

- *Responder 2.* Rejecting makes her the proposer next period, worth $v-a-c=b$. Accepting at least b and rejecting less is optimal (indifferent at b). Symmetrically for responder 1, whose rejection is worth $v-b-c=a$.
- *Proposer 1.* Offering b yields $v-b-c=a$; offering more yields less; an offer below b is rejected and yields $-c+a<a$.
- Infinite disagreement gives each player $-\infty$ (infinitely many offers), so a profitable deviation would have to end in agreement after finitely many periods, and the one-shot checks above suffice.

So *every* $\sigma(a)$ is an SPNE, with immediate agreement and net payoffs

$$(u_1, u_2) = (a,\ v-c-a), \qquad a \in [0,\ v-c].$$

Indifference only requires $a+b=v-c$, one equation in two unknowns. With discounting it was two independent equations, $b=\delta_2(v-a)$ and $a=\delta_1(v-b)$.

Why Shaked–Sutton fails here. Let M_i, m_i be the sup and inf of player i's SPNE payoffs as proposer. The argument of Example 9.AA.2 gives $m_1 \geq v-c-M_2$, $M_1 \leq v-c-m_2$, and symmetrically for player 2. Substituting, $M_1 \leq v-c-(v-c-M_1) = M_1$: the bound is an identity. With discounting the same step gives $M_1 \leq v-\delta_2(v-\delta_1 M_1)$, a contraction with slope $\delta_1\delta_2<1$ and a unique fixed point. Here the slope is 1 and every point is a fixed point. The bounds still show that every SPNE gives the first proposer a payoff in $[0, v-c]$ (since $m_1 \geq v-c-M_2 \geq 0$), and the stationary equilibria fill that whole interval, so *any* payoff in $[0, v-c]$ for player 1 is an SPNE outcome.

Equilibria with delay. Non-stationary strategies also support delay when $c \leq v/2$. In period 1 player 1 offers player 2 nothing and she rejects; from period 2 on $\sigma(v-c)$ is played (player 2 offers player 1 $v-c$, keeping c). If player 1 instead offers anything else in period 1 and it is rejected, $\sigma(0)$ is played from period 2. Player 2 rejecting the equilibrium offer gets $c-c=0$, the same as accepting nothing (a weak best response). After a deviant offer her rejection is worth $v-c$, so player 1 must offer her at least $v-c$ and keeps at most $c-c=0$, or gets $-c$ if the offer is rejected. On the path player 1 gets $-c+(v-c)=v-2c \geq 0$. Total net surplus is $v-2c<v-c$: an extra offer cost is wasted, so SPNE outcomes need not even be efficient.

Interpretation. How a bargaining game is specified matters (MWG p. 299 names Exercises 9.B.7 and 9.B.13 as the illustration). Discounting makes delay costly *in proportion to what is at stake*, which delivers a unique split that is the limit of the finite games. A fixed cost per offer makes the last-mover advantage absolute in finite games and leaves the infinite game indeterminate: any division of $v-c$, and even inefficient delay, can be sustained by expectations about who will concede. Bargaining power then comes from expectations and conventions, not from the procedure. This is a warning against reading Rubinstein's uniqueness as a general law of bargaining (compare M&R Ch. 5 on the traditional view that small-numbers bargaining is indeterminate).

Verification. `check_D.py` runs the backward-induction recursion for $T=1,\ldots,11$ with $(v,c)=(10,3)$: $(u_1,u_2)=(7,0)$ with offer $(10,0)$ for odd T, and $(0,7)$ with offer $(3,7)$ for even T, matching

$(v-c)[1-(-1)^T]/2$. It checks that the two indifference conditions of the infinite game are the same equation, that the Shaked–Sutton map has slope 1 here versus $\delta_1\delta_2$ with discounting, and the delay equilibrium's payoffs. The MWG solutions manual agrees on both parts: for 9.B.7 it gives $(v,0)$ in period 1 when T is odd and $(c,v-c)$ when T is even, and for 9.B.13 it states that there are many SPNE and refers to Rubinstein (1982).

Answer. Finite T (9.B.7): unique SPNE with immediate agreement. Player 1 proposes $(v,0)$ when T is odd (net payoffs $(v-c,0)$) and $(c,v-c)$ when T is even (net $(0,v-c)$), so the last proposer takes everything net of one offer cost and there is no limit as $T\to\infty$. Infinite horizon (9.B.13): no uniqueness. Each $a\in[0,v-c]$ gives a stationary SPNE with immediate split $(a,v-c-a)$, and for $c\le v/2$ there are SPNE with delay. The Shaked–Sutton contraction has slope 1 instead of $\delta_1\delta_2$.

29 MWG Exercise 9.B.12

Advanced · block B2

Source: MWG Exercise 9.B.12, page 302–303 (statement in the book; not reproduced here).

What is asked. In the infinite-horizon alternating-offer game of MWG Example 9.AA.2, let the two players discount the future with different factors $\delta_1\neq\delta_2$, and find the (unique) subgame perfect Nash equilibrium.

Method. Rubinstein (1982) as presented in MWG Appendix A (Example 9.AA.2, p. 298): guess stationary strategies from the two indifference conditions, verify them with one-shot deviations, and prove uniqueness with the Shaked–Sutton bounds on the sup and inf of SPNE payoffs. Notation as in the course notes (Part II, "Infinite horizon: Rubinstein"), with pie v.

Solution. (a) Stationary strategies. Let x_i be what player i keeps when proposing. A responder j who rejects becomes the proposer next period and gets x_j then, worth $\delta_j x_j$ today. The proposer offers exactly that:

$$v-x_1=\delta_2 x_2,\qquad v-x_2=\delta_1 x_1.$$

Solving,

$$\boxed{x_1^*=\frac{1-\delta_2}{1-\delta_1\delta_2}\,v,\qquad x_2^*=\frac{1-\delta_1}{1-\delta_1\delta_2}\,v.}$$

SPNE strategies: player i, when proposing, keeps x_i^* and offers $v-x_i^*=\delta_j x_j^*$; when responding, accepts iff offered at least $\delta_i x_i^*$. On the path player 1 proposes in period 1 and player 2 accepts:

$$u_1=x_1^*=\frac{1-\delta_2}{1-\delta_1\delta_2}\,v,\qquad u_2=v-x_1^*=\frac{\delta_2(1-\delta_1)}{1-\delta_1\delta_2}\,v.$$

Verification (one-shot deviations, valid with discounting). A responder j offered $y\ge\delta_j x_j^*$ gets y by accepting and $\delta_j x_j^*$ by rejecting, so accepting is optimal; below that, rejecting is optimal. A proposer i who offers more than $\delta_j x_j^*$ gives away surplus; one who offers less is rejected and then receives $\delta_i x_i^*$ one period later, worth $\delta_i\cdot\delta_i x_i^*<x_i^*$ today.

(b) Uniqueness (Shaked–Sutton). Let M_i and m_i be the sup and inf of player i's SPNE payoffs in subgames where i proposes; by stationarity they do not depend on the date.

- Player 2 accepts anything above $\delta_2 M_2$ and rejects anything below $\delta_2 m_2$. Hence $m_1\ge v-\delta_2 M_2$. A rejected offer is no better, because the pie is only available later, so $M_1\le v-\delta_2 m_2$.

- Symmetrically, $m_2 \geq v - \delta_1 M_1$ and $M_2 \leq v - \delta_1 m_1$.

Combining, $M_1 \leq v - \delta_2(v - \delta_1 M_1)$, so $M_1 \leq (1-\delta_2)v/(1-\delta_1\delta_2)$; and $m_1 \geq v - \delta_2(v - \delta_1 m_1)$, so $m_1 \geq (1-\delta_2)v/(1-\delta_1\delta_2)$. Hence $M_1 = m_1 = x_1^*$, and likewise $M_2 = m_2 = x_2^*$. Since every SPNE gives the first proposer exactly x_1^*, and any delay would shrink the total below v and leave someone worse off than these bounds allow, agreement is immediate and the strategies in (a) are the unique SPNE. The key step is that the map $M \mapsto v - \delta_2(v - \delta_1 M)$ is a contraction with slope $\delta_1\delta_2 < 1$.

(c) Comparative statics and limits.

- *Equal factors*: $\delta_1 = \delta_2 = \delta$ gives $x_1^* = v/(1+\delta)$, the result of Example 9.AA.2 and the $T \to \infty$ limit of Example 9.AA.1.
- *Patience is power*: $\partial x_1^*/\partial \delta_1 = \delta_2(1-\delta_2)v/(1-\delta_1\delta_2)^2 > 0$, and x_1^* falls in δ_2.
- $\delta_1 \to 1$ with $\delta_2 < 1$ fixed: $x_1^* \to v$, so the perfectly patient player takes everything. With $\delta_2 \to 1$ and $\delta_1 < 1$ fixed: $x_1^* \to 0$, so player 2 gets everything even though player 1 moves first.
- *Frequent offers*: with $\delta_i = e^{-r_i\Delta}$ and Δ the time between offers, $1 - e^{-r_2\Delta} \approx r_2\Delta$ and $1 - e^{-(r_1+r_2)\Delta} \approx (r_1+r_2)\Delta$, so
$$\lim_{\Delta \to 0} x_1^* = \frac{r_2}{r_1 + r_2}\, v.$$
The first-mover advantage disappears and only relative impatience matters; equal rates give $v/2$.
- *Example*: $\delta_1 = 0.9$, $\delta_2 = 0.8$, $v = 1$: $x_1^* = 0.2/0.28 = 5/7 \approx 0.714$, player 2 gets $2/7$; if player 2 proposed first she would keep $x_2^* = 0.1/0.28 = 5/14$. With $r_1 = 1$, $r_2 = 2$ and $\Delta \to 0$ player 1 gets $2/3$.

Interpretation. Under complete information bargaining is efficient (immediate agreement) and the split is determinate: it rewards the player who can better afford to wait. That is the formal content of "bargaining power" in the transfers that the Coase theorem and M&R's value maximization principle leave open; discount factors move the split, never the value-creating decision. Contrast item 28, where a fixed per-offer cost instead of discounting destroys the uniqueness.

Verification. `check_D.py` confirms symbolically that x_1^*, x_2^* solve both indifference conditions and the Shaked–Sutton fixed point, the equal-δ case, both one-sided limits, the $\Delta \to 0$ limit $r_2/(r_1+r_2)$, and the derivative. It also runs the finite-horizon recursion of the course notes with $(\delta_1, \delta_2) = (0.9, 0.8)$ for $T = 400$ and 401, which gives $5/7$ to machine precision: the infinite-horizon split is the limit of the finite ones. The MWG solutions manual gives the same strategies: the proposer i offers j the share $\delta_j(1-\delta_i)v/(1-\delta_i\delta_j)$, and j accepts iff offered at least that.

Answer. Unique SPNE: player i as proposer keeps $x_i^* = (1-\delta_j)v/(1-\delta_1\delta_2)$ and offers $\delta_j x_j^*$; as responder accepts iff offered at least $\delta_i x_i^*$. Agreement is immediate with payoffs $\big((1-\delta_2)v,\ \delta_2(1-\delta_1)v\big)/(1-\delta_1\delta_2)$. The share rises in own patience; $\delta_1 \to 1$ gives player 1 everything; with $\delta_i = e^{-r_i\Delta}$ and $\Delta \to 0$, player 1 gets $r_2 v/(r_1+r_2)$.

30 Cabral 13.10 Asset specificity and vertical integration Advanced · block B3

Source: Cabral 13.10 Asset specificity and vertical integration, page 346 (statement in the book; not reproduced here).

What is asked. Two firms M and R invest e_M, e_R (in \$) in a joint project worth $v = \sqrt{e_M}+\sqrt{e_R}$ and split v equally after investing. Find (a) the equilibrium investments, (b) the investments chosen by the merged firm, (c) total payoffs under separation and integration, and (d) comment in light of §13.1, Box 13.1 and §5.1.

Method. Simultaneous investment game with ex-post bargaining, solved by first-order conditions; joint-surplus maximization for the merged firm. Hold-up and specific assets (Cabral §13.1, "Investment incentives", pp. 338–339; §3.4, p. 55); double marginalization and Box 13.1 (pp. 334–336); externalities (§5.1, pp. 94–96). Course link: the hold-up preview in the Aula 2 notes and M&R Ch. 5, investments and specific assets (pp. 134–139).

Solution.

(a) Vertical separation. Investment is sunk before the split, and the split is fixed at 50/50 whatever was invested. Firm M solves

$$\max_{e_M \geq 0} \ \tfrac{1}{2}\big(\sqrt{e_M} + \sqrt{e_R}\big) - e_M.$$

The objective is strictly concave in e_M; the first-order condition is

$$\frac{1}{4\sqrt{e_M}} = 1 \implies \sqrt{e_M} = \tfrac{1}{4} \implies e_M^S = \tfrac{1}{16}.$$

M's choice does not depend on e_R (a dominant strategy), and by symmetry $e_R^S = \frac{1}{16}$. Then $v^S = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, and each firm earns $\frac{1}{2} \cdot \frac{1}{2} - \frac{1}{16} = \frac{3}{16}$.

(b) Vertical integration. Firm $M\&R$ keeps all of v and pays both costs:

$$\max_{e_M, e_R \geq 0} \ \sqrt{e_M} + \sqrt{e_R} - e_M - e_R \quad \implies \quad \frac{1}{2\sqrt{e_i}} = 1 \quad \implies \quad e_M^I = e_R^I = \tfrac{1}{4}.$$

Then $v^I = \frac{1}{2} + \frac{1}{2} = 1$. Each investment is four times the separated level.

(c) Total payoffs.

$$W^S = v^S - e_M^S - e_R^S = \tfrac{1}{2} - \tfrac{1}{8} = \tfrac{3}{8}, \qquad W^I = v^I - e_M^I - e_R^I = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

Integration raises total payoff by $\frac{1}{2} - \frac{3}{8} = \frac{1}{8}$, a gain of one third over separation (a quarter of the efficient surplus is lost under separation).

	e_M	e_R	v	Total payoff
Separation, 50/50 split	1/16	1/16	1/2	3/8
Integration	1/4	1/4	1	1/2

(d) Comment. *Hold-up (§13.1).* Each investment is specific: it creates value only inside the relationship, and once sunk it no longer affects how v is divided. Each firm pays the full marginal cost of investing but receives only half the marginal value, so it equates $\frac{1}{2} \cdot \frac{1}{2\sqrt{e_i}}$ with 1 instead of $\frac{1}{2\sqrt{e_i}}$ with 1. This is the continuous version of Cabral's hold-up example (p. 338): there, an investment worth 8 and costing 5 was not made because the investor would get only 4 in an even split (p. 339); here investment is made, but at a quarter of the efficient level. The margin note on p. 339 says this exercise is "the case of a continuum of choices". Integration puts both decisions under one owner who receives the whole return, which removes the distortion; Cabral adds that it creates an agency problem in its place, since the acquired unit's manager now follows orders with weaker incentives (pp. 338–339). Integration is worth it only if that agency cost is below the gain $\frac{1}{8}$.

Externalities (§5.1). The 50/50 split makes each firm's investment confer a positive externality on the other: half of the extra value goes to the partner, who did not pay for it. As with any positive externality (front yards, item 9), the activity is under-provided. The Coase remedy (bargain to the efficient level) would require contracting on e_M and e_R *before* they are sunk, which is exactly what is assumed impossible: investments are hard to specify and verify, so contracts are incomplete (p. 338). Ex-post bargaining is efficient given the investments; it is the ex-ante investments that are distorted.

Box 13.1 and double marginalization. Box 13.1 (p. 336) reports that integrated distributors and sports networks charge consumers lower prices, evidence that integration removes the double-marginalization externality: under separation each firm adds its own margin and ignores the effect on the other's profit (pp. 334–336). The hold-up problem is the investment counterpart of the same failure: separated firms ignore the effect of their choice on the partner's payoff, and integration internalizes it in both cases.

Interpretation. The exercise isolates why ownership matters even when bargaining after the fact is efficient: the anticipated split determines the incentive to invest beforehand. This is the core of the transaction-cost and property-rights theories of the firm and the Topic 6 material that the Aula 2 notes preview (their quadratic example gives the same $\frac{3}{8}$ against $\frac{1}{2}$). It also explains the Smart-car strain of item 13: a jointly owned venture with investments by both partners is a 50/50 split.

Verification. (1) `check_B.py` confirms with sympy: $e_i^S = \frac{1}{16}$, payoff $\frac{3}{16}$ each, $W^S = \frac{3}{8}$; $e_i^I = \frac{1}{4}$, $W^I = \frac{1}{2}$; difference $\frac{1}{8}$, a one-third gain. (2) *No split fixes it.* If M receives a share s and R receives $1 - s$, then $e_M = s^2/4$, $e_R = (1-s)^2/4$ and $W(s) = \frac{1}{2} - \frac{1}{4}\left[s^2 + (1-s)^2\right]$, maximized at $s = \frac{1}{2}$ with $W = \frac{3}{8} < \frac{1}{2}$; giving everything to one firm ($s = 1$) yields only $\frac{1}{4}$, because the other then stops investing. With two-sided investment, no ex-post division restores efficiency; joint ownership of both decisions does. (3) *One-sided case:* if only M invested, giving M the whole value ($s = 1$) would induce the efficient $e_M = \frac{1}{4}$, so ownership should go to the investor. That is the property-rights logic behind the exercise.

Answer. (a) $e_M^S = e_R^S = 1/16$ (from $\frac{1}{4\sqrt{e_i}} = 1$), payoff 3/16 each. (b) $e_M^I = e_R^I = 1/4$. (c) $W^S = 3/8 < W^I = 1/2$: integration gains 1/8 (one third). (d) A 50/50 ex-post split gives each investor half the marginal value of its sunk, specific investment, a positive externality on the partner and a continuous hold-up problem (§13.1, §5.1). Integration internalizes it, as it internalizes double marginalization (Box 13.1), at the cost of new agency problems.